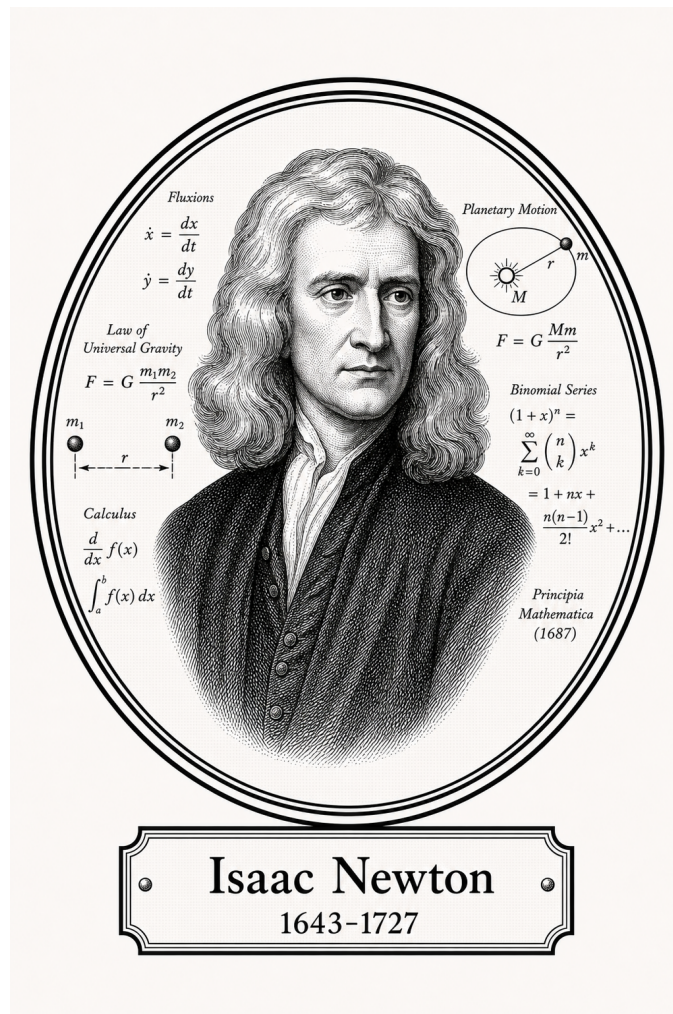


FROM TURING TO CARMACK

# Applied and Computational Mathematics



Isaac Newton

1643-1727

*Self-Study Notes*

William Sollers

July 1, 2026

*For every reader learning to make mathematics their own.*

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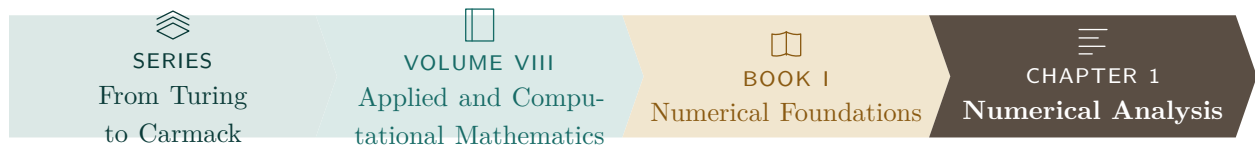
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Part I

Applied and Computational  
Mathematics

# Chapter 1

## Numerical Analysis



### 1.1 IEEE 754 Floating-Point Formats

#### Toolkit — Floating-Point Representation

This section develops the structural language of IEEE 754 binary floating-point representation. The key ideas are:

- every binary floating-point datum is organized into sign, exponent, and fraction fields,
- the exponent field determines whether the encoding is normalized, subnormal, zero, infinity, or NaN,
- the fraction field determines the significant bits of the represented value,
- each standard format is determined by its total bit width together with its exponent width, fraction width, and exponent bias.

## IEEE 754 Floating-Point Formats

### Definition - IEEE 754 Binary Floating-Point Datum

**Definition 1.1.1** (IEEE 754 Binary Floating-Point Datum). An *IEEE 754 binary floating-point datum* is a binary encoding partitioned into three fields:

- a *sign field* consisting of one bit,
- an *exponent field* consisting of a fixed number of bits,
- a *fraction field* consisting of a fixed number of bits.

Its interpretation depends on the exponent-field pattern and the fraction-field pattern.

*Remark* (Standard quantified statement).

$\text{IEEEBinaryFloatDatum}(x) \iff \text{HasSignField}(x) \wedge \text{HasExponentField}(x) \wedge \text{HasFractionField}(x).$  ■

*Remark* (Interpretation). IEEE 754 does not interpret a floating-point bit string as a plain binary integer. Instead, the sign field determines sign, the exponent field determines scale and special cases, and the fraction field records the significant bits of the encoded value.

### Definition - Normalized IEEE 754 Binary Number

**Definition 1.1.2** (Normalized IEEE 754 Binary Number). Let an IEEE 754 binary floating-point format have exponent bias  $\text{bias}$ . A finite encoded datum is called *normalized* if its exponent field is neither all zeros nor all ones.

If  $s$  is the sign bit, if  $e$  is the unsigned integer determined by the exponent field, and if  $f$  is the binary fraction determined by the fraction field, then the represented value is

$$(-1)^s (1.f)_2 2^{e-\text{bias}}.$$

*Remark* (Standard quantified statement).

$\text{NormalizedIEEEBinaryNumber}(x) \iff \text{IEEEBinaryFloatDatum}(x) \wedge \text{ExponentFieldNeitherAllZerosNorAllOnes}(x).$

*Remark* (Interpretation). In a normalized encoding, the leading binary digit of the significand is implicitly 1. This hidden leading bit is what gives normalized numbers one more bit of effective precision than the stored fraction field alone.

### Definition - Subnormal IEEE 754 Binary Number

**Definition 1.1.3** (Subnormal IEEE 754 Binary Number). Let an IEEE 754 binary floating-point format have exponent bias  $\text{bias}$ . A finite encoded datum is called *subnormal* if its exponent field is all zeros and its fraction field is not all zeros.

If  $s$  is the sign bit and if  $f$  is the binary fraction determined by the fraction field, then the represented value is

$$(-1)^s (0.f)_2 2^{1-\text{bias}}.$$

*Remark* (Standard quantified statement).

$\text{SubnormalIEEEBinaryNumber}(x) \iff \text{IEEEBinaryFloatDatum}(x) \wedge \text{ExponentFieldAllZeros}(x) \wedge \neg \text{FractionFieldAllZeros}(x)$

*Remark* (Interpretation). Subnormal numbers fill the gap between zero and the smallest positive normalized number. They preserve gradual underflow by allowing the leading significand bit to be 0 instead of the implicit 1 used for normalized numbers.

#### Definition - IEEE 754 Infinity Encoding

**Definition 1.1.4** (IEEE 754 Infinity Encoding). An IEEE 754 binary floating-point datum encodes *infinity* if its exponent field is all ones and its fraction field is all zeros. The sign bit determines whether the represented value is  $+\infty$  or  $-\infty$ .

*Remark* (Standard quantified statement).

$\text{IEEEInfinityEncoding}(x) \iff \text{IEEEBinaryFloatDatum}(x) \wedge \text{ExponentFieldAllOnes}(x) \wedge \text{FractionFieldAllZeros}(x)$

*Remark* (Interpretation). Infinity is a distinguished non-finite encoding used to represent overflow and certain limit-like arithmetic results. The sign field still matters, so IEEE 754 distinguishes positive infinity from negative infinity.

#### Definition - IEEE 754 NaN Encoding

*Remark* (Dependencies).

- [IEEE 754 Binary Floating-Point Datum](#)

**Definition 1.1.5** (IEEE 754 NaN Encoding). An IEEE 754 binary floating-point datum encodes *NaN* (Not a Number) if its exponent field is all ones and its fraction field is not all zeros.

*Remark* (Standard quantified statement).

$\text{IEEENaNEncoding}(x) \iff \text{IEEEBinaryFloatDatum}(x) \wedge \text{ExponentFieldAllOnes}(x) \wedge \neg \text{FractionFieldAllZeros}(x)$

*Remark* (Interpretation). A NaN encoding represents an undefined or invalid numerical result rather than a real number. For example, NaNs arise from operations such as invalid indeterminate forms or from the propagation of an already invalid datum through later computations.

Definition - IEEE 754 Binary16 Format

**Definition 1.1.6** (IEEE 754 Binary16 Format). The *IEEE 754 binary16 format* is the 16-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 5 exponent bits,
- 10 fraction bits,
- exponent bias 15.

For a normalized finite binary16 datum, the represented value is

$$(-1)^s(1.f)_22^{e-15}.$$

*Remark* (Standard quantified statement).

$$\text{Binary16}(x) \iff \text{IEEEBinaryFloatDatum}(x) \wedge \text{BitWidth}(x, 16) \wedge \text{SignWidth}(x, 1) \wedge \text{ExponentWidth}(x, 5) \wedge \text{FractionWidth}(x, 10)$$

*Remark* (Interpretation). Binary16 is commonly called *half precision*. It is useful when reduced storage and bandwidth matter more than wide dynamic range or high precision.

*Remark* (Bit layout).

|     |          |    |   |          |
|-----|----------|----|---|----------|
| 15  | 14       | 10 | 9 | 0        |
| $s$ | exponent |    |   | fraction |

Definition - IEEE 754 Binary32 Format

**Definition 1.1.7** (IEEE 754 Binary32 Format). The *IEEE 754 binary32 format* is the 32-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 8 exponent bits,
- 23 fraction bits,
- exponent bias 127.

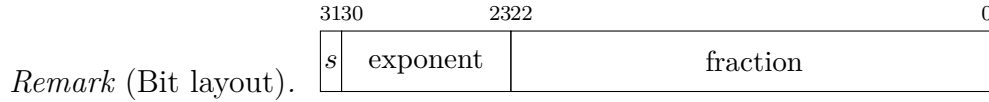
For a normalized finite binary32 datum, the represented value is

$$(-1)^s(1.f)_22^{e-127}.$$

*Remark* (Standard quantified statement).

$$\text{Binary32}(x) \iff \text{IEEEBinaryFloatDatum}(x) \wedge \text{BitWidth}(x, 32) \wedge \text{SignWidth}(x, 1) \wedge \text{ExponentWidth}(x, 8) \wedge \text{FractionWidth}(x, 23)$$

*Remark* (Interpretation). Binary32 is commonly called *single precision*. It is the standard 32-bit floating-point format used broadly in scientific computing, computer graphics, simulation, and machine learning.



#### Definition - IEEE 754 Binary64 Format

**Definition 1.1.8** (IEEE 754 Binary64 Format). The *IEEE 754 binary64 format* is the 64-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 11 exponent bits,
- 52 fraction bits,
- exponent bias 1023.

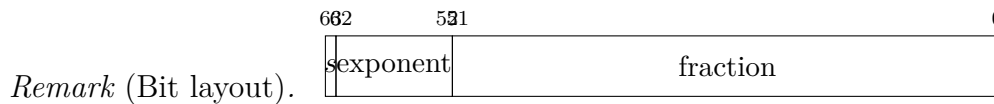
For a normalized finite binary64 datum, the represented value is

$$(-1)^s(1.f)_22^{e-1023}.$$

Remark (Standard quantified statement).

$\text{Binary64}(x) \iff \text{IEEEBinaryFloatDatum}(x) \wedge \text{BitWidth}(x, 64) \wedge \text{SignWidth}(x, 1) \wedge \text{ExponentWidth}(x, 11) \wedge \text{FractionWidth}(x, 52)$

Remark (Interpretation). Binary64 is commonly called *double precision*. It is the standard high-precision floating-point format for much of scientific and engineering computation.



#### Definition - IEEE 754 Binary128 Format

**Definition 1.1.9** (IEEE 754 Binary128 Format). The *IEEE 754 binary128 format* is the 128-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 15 exponent bits,
- 112 fraction bits,
- exponent bias 16383.

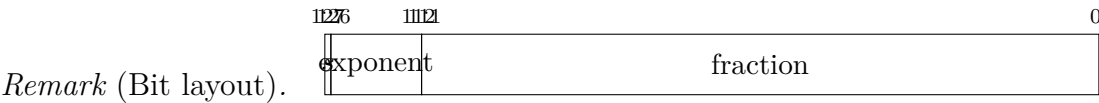
For a normalized finite binary128 datum, the represented value is

$$(-1)^s(1.f)_22^{e-16383}.$$

Remark (Standard quantified statement).

$\text{Binary128}(x) \iff \text{IEEEBinaryFloatDatum}(x) \wedge \text{BitWidth}(x, 128) \wedge \text{SignWidth}(x, 1) \wedge \text{ExponentWidth}(x, 15) \wedge \text{FractionWidth}(x, 112)$

*Remark* (Interpretation). Binary128 is commonly called *quadruple precision*. It provides substantially greater range and precision than binary64 at substantially greater storage and computational cost.



*Remark* (Summary). The four standard IEEE 754 basic binary interchange formats considered here are:

binary16 (1, 5, 10),      binary32 (1, 8, 23),      binary64 (1, 11, 52),      binary128 (1, 15, 112),

where each triple records

(sign bits, exponent bits, fraction bits).

## 1.2 Interval Arithmetic

### Box (Interval) Arithmetic in $\mathbb{R}^n$ - Quick Reference

| Operation                         | Rule                                                                            |
|-----------------------------------|---------------------------------------------------------------------------------|
| Box (product interval)            | $[a, b] := \{x \in \mathbb{R}^n : a \leq x \leq b\}$                            |
| Containment                       | $[a, b] \subseteq [c, d] \iff c \leq a \wedge b \leq d$                         |
| Minkowski sum                     | $[a, b] \oplus [c, d] = [a + c, b + d]$                                         |
| Difference                        | $[a, b] \ominus [c, d] = [a - d, b - c]$                                        |
| Scalar mult. ( $\lambda \geq 0$ ) | $\lambda[a, b] = [\lambda a, \lambda b]$                                        |
| Scalar mult. ( $\lambda < 0$ )    | $\lambda[a, b] = [\lambda b, \lambda a]$                                        |
| Coordinatewise product            | $[a, b] \odot [c, d] = \prod_i [\min S_i, \max S_i]$                            |
| Norm bound (Euclidean)            | $\ x\ _2 \leq \max\{\ a\ _2, \ b\ _2\}$ for $x \in [a, b]$ if $0 \leq a \leq b$ |

### Interval Arithmetic - Quick Reference

| Operation      | Rule                                              |
|----------------|---------------------------------------------------|
| Addition       | $[a, b] + [c, d] = [a + c, b + d]$                |
| Subtraction    | $[a, b] - [c, d] = [a - d, b - c]$                |
| Multiplication | $[a, b] \cdot [c, d]$ (case analysis)             |
| Containment    | $[a, b] \subseteq [c, d] \iff c \leq a, b \leq d$ |

### 1.2.1 Interval Arithmetic

#### Definition (Interval Addition)

**Definition 1.2.1** (Interval Addition). For closed intervals  $[a, b]$  and  $[c, d]$ , their *sum* is

$$[a, b] + [c, d] = [a + c, b + d].$$

*Remark* (Dependencies).

- [Closed Interval](#)
- [Addition on  \$\mathbb{R}\$](#)

#### Definition (Interval Subtraction)

**Definition 1.2.2** (Interval Subtraction). For closed intervals  $[a, b]$  and  $[c, d]$ , their *difference* is

$$[a, b] - [c, d] = [a - d, b - c].$$

*Remark* (Dependencies).

- [Closed Interval](#)
- [Subtraction on  \$\mathbb{R}\$](#)

#### Definition (Interval Multiplication)

**Definition 1.2.3** (Interval Multiplication). For closed intervals  $[a, b]$  and  $[c, d]$ , their *product* is

$$[a, b] \cdot [c, d] = [\min(ac, ad, bc, bd), \max(ac, ad, bc, bd)].$$

*Remark* (Dependencies).

- [Closed Interval](#)
- [Multiplication on  \$\mathbb{R}\$](#)
- [Order on  \$\mathbb{R}\$](#)

#### Definition (Interval Containment)

**Definition 1.2.4** (Interval Containment).  $[a, b] \subseteq [c, d]$  if and only if  $c \leq a$  and  $b \leq d$ .

*Remark* (Dependencies).

- [Closed Interval](#)
- [Subset](#)
- [Order on  \$\mathbb{R}\$](#)

*Remark* (Motivation). Interval arithmetic provides a framework for rigorous numerical computation with guaranteed error bounds. A real number  $x \in [a, b]$  means the computed value may lie anywhere in the interval; operations propagate these bounds.

*Remark* (Status). *This section is a stub. A full treatment – the sub-distributivity law, dependency problem, applications to root-finding, and connections to compactness – will be developed in a future revision.*

## 1.2.2 Interval Arithmetic in $\mathbb{R}^n$

### Definition (Box / Product Interval)

**Definition 1.2.5** (Box / Product Interval). Let  $a, b \in \mathbb{R}^n$  with  $a \leq b$  coordinatewise (i.e.,  $a_i \leq b_i$  for all  $i$ ). The *box* (or *interval vector*) determined by  $(a, b)$  is

$$[a, b] := \{x \in \mathbb{R}^n : a \leq x \leq b\} = \prod_{i=1}^n [a_i, b_i].$$

*Remark* (Dependencies).

- Cartesian Product
- Closed Interval
- Order on  $\mathbb{R}$

### Definition (Box Containment)

**Definition 1.2.6** (Box Containment). For  $a \leq b$  and  $c \leq d$  in  $\mathbb{R}^n$ , we write  $[a, b] \subseteq [c, d]$  iff

$$(\forall x \in \mathbb{R}^n) (x \in [a, b] \Rightarrow x \in [c, d]).$$

Equivalently (and most useful computationally),

$$[a, b] \subseteq [c, d] \iff c \leq a \wedge b \leq d \quad (\text{coordinatewise}).$$

*Remark* (Dependencies).

- Box / Product Interval
- Subset
- Order on  $\mathbb{R}$

### Definition (Minkowski Addition of Boxes)

**Definition 1.2.7** (Minkowski Addition of Boxes). For boxes  $X = [a, b]$  and  $Y = [c, d]$  in  $\mathbb{R}^n$ , their *Minkowski sum* is

$$X \oplus Y := \{x + y : x \in X, y \in Y\}.$$

For boxes, this closes in the class of boxes and satisfies the endpoint rule

$$[a, b] \oplus [c, d] = [a + c, b + d].$$

*Remark* (Dependencies).

- Box / Product Interval

- [Addition on  \$\mathbb{R}\$](#)

#### Definition (Box Subtraction / Difference)

**Definition 1.2.8** (Box Subtraction / Difference). For boxes  $X = [a, b]$  and  $Y = [c, d]$  in  $\mathbb{R}^n$ , define

$$X \ominus Y := \{x - y : x \in X, y \in Y\}.$$

Then

$$[a, b] \ominus [c, d] = [a - d, b - c].$$

*Remark* (Dependencies).

- [Box / Product Interval](#)
- [Subtraction on  \$\mathbb{R}\$](#)

#### Definition (Scalar Multiplication of a Box)

**Definition 1.2.9** (Scalar Multiplication of a Box). For  $\lambda \in \mathbb{R}$  and a box  $[a, b] \subseteq \mathbb{R}^n$ , define

$$\lambda[a, b] := \{\lambda x : x \in [a, b]\}.$$

Then

$$\lambda[a, b] = \begin{cases} [\lambda a, \lambda b], & \lambda \geq 0, \\ [\lambda b, \lambda a], & \lambda < 0, \end{cases}$$

where the inequalities are coordinatewise.

*Remark* (Dependencies).

- [Box / Product Interval](#)
- [Multiplication on  \$\mathbb{R}\$](#)
- [Order on  \$\mathbb{R}\$](#)

#### Definition (Coordinatewise / Hadamard Product of Boxes)

**Definition 1.2.10** (Coordinatewise / Hadamard Product of Boxes). For  $x, y \in \mathbb{R}^n$ , write  $(x \odot y)_i := x_i y_i$ . For boxes  $X = [a, b]$  and  $Y = [c, d]$ , define

$$X \odot Y := \{x \odot y : x \in X, y \in Y\}.$$

Then  $X \odot Y$  is again a box, computed coordinatewise:

$$[a, b] \odot [c, d] = \prod_{i=1}^n [\min S_i, \max S_i], \quad S_i := \{a_i c_i, a_i d_i, b_i c_i, b_i d_i\}.$$

*Remark* (Dependencies).

- [Box / Product Interval](#)
- [Multiplication on  \$\mathbb{R}\$](#)
- [Order on  \$\mathbb{R}\$](#)

*Remark* (Interpretation: intervals as guaranteed enclosures). A box  $X = [a, b] \subseteq \mathbb{R}^n$  represents a *set of possible vectors* (e.g., a computed vector plus uncertainty bounds). The operations above are *set operations* (Minkowski sum/difference, scalar image), so the result is a *guaranteed enclosure* of all possible true outcomes under the corresponding vector operation.

*Remark* (Norm bounds and why boxes are convenient). For many applications you also want quick bounds on a norm over a box. For example, if  $0 \leq a \leq b$  coordinatewise, then for every  $x \in [a, b]$  we have  $0 \leq x \leq b$  and hence  $\|x\|_2 \leq \|b\|_2$ . More generally, bounding norms over a box can be reduced to checking finitely many “corners” when the objective is coordinatewise monotone.

*Remark* (Status). *This section is a stub. A full treatment – interval extensions of general functions  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ , the dependency problem, sub-distributivity, natural vs. centered (affine) forms, and connections to compactness and continuity – will be developed in a future revision.*

## 1.3 Numerical Geometry

### 1.3.1 Week 1: Points, Vectors, and Floating-Point Geometry

Reading checklist.

- Atkinson, Chapter 1
- Solomon, Chapter 1 selectively
- Solomon, Chapter 2 selectively
- Vince, Chapter 7
- Existing Volume VII IEEE 754 notes
- Existing Volume VII interval arithmetic notes

*Remark* (Project placeholder). `lra-numerical-analysis/projects/point-vector-01`

*Remark* (Handwritten notes placeholder). TODO: Add handwritten-note extraction here after study.

## 1.3.2 Week 2: Linear Interpolation and Affine Sampling

Reading checklist.

- Salomon, Chapter 1 sections 1.1–1.3
- Salomon, Chapter 2 section 2.1
- Vince, Chapter 13
- Piegl–Tiller, Chapter 1 skim only
- Piegl–Tiller, Chapter 2 preview only

*Remark* (Project placeholder). `lra-numerical-analysis/projects/linear-interpolation-02`

*Remark* (Handwritten notes placeholder). TODO: Add handwritten-note extraction here after study.

## 1.3.3 Projects

- `point-vector-01`: planned
- `linear-interpolation-02`: planned
- `bezier-curve-03`: deferred
- `bspline-basis-04`: deferred
- `nurbs-curve-05`: deferred

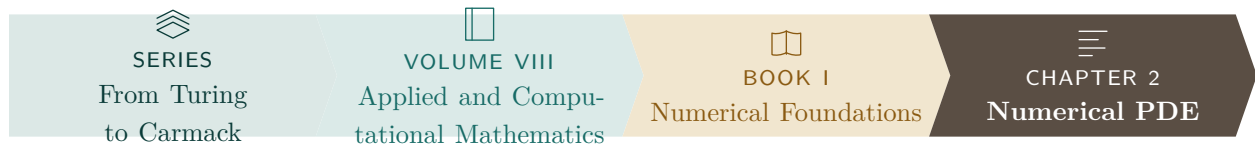
## 1.3.4 Exercises

- Week 1 exercises: planned
- Week 2 exercises: planned
- TODO: Add exercises after handwritten notes are extracted.

# Proofs

# Chapter 2

## Numerical PDE



### 2.1 Numerical PDE

*Remark* (Planned content). Active mathematical content has not yet been added.

### Proofs

# Chapter 3

## Calculus



### 3.1 Limits

#### 3.1.1 Limits

##### Definitions

**Definition** (Intuitive Limit Idea). The limit  $\lim_{x \rightarrow a} f(x) = L$  says that the values of  $f(x)$  can be made close to  $L$  by taking  $x$  close to  $a$ , without requiring  $f(a)$  itself to be defined or equal to  $L$ .

**Definition** (Epsilon-Delta Reference Definition). The statement  $\lim_{x \rightarrow a} f(x) = L$  means that for every  $\varepsilon > 0$  there is a  $\delta > 0$  such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

In Volume VI this definition is mainly a reference point for interpreting limit notation; detailed proofs belong in Real Analysis.

**Definition** (One-Sided Limits). The right-hand limit  $\lim_{x \rightarrow a^+} f(x)$  tracks values with  $x > a$ . The left-hand limit  $\lim_{x \rightarrow a^-} f(x)$  tracks values with  $x < a$ . The two-sided limit exists exactly when both one-sided limits exist and agree.

**Definition** (Limits at Infinity). The notation  $\lim_{x \rightarrow \infty} f(x) = L$  describes the value approached by  $f(x)$  as  $x$  becomes arbitrarily large. The notation  $\lim_{x \rightarrow a} f(x) = \infty$  describes unbounded growth near  $a$ .

**Definition** (Indeterminate Forms). Forms such as  $0/0$ ,  $\infty/\infty$ ,  $0 \cdot \infty$ ,  $\infty - \infty$ ,  $1^\infty$ ,  $0^0$ , and  $\infty^0$  do not determine a limit by inspection. They signal that algebraic or analytic rewriting is needed.

**Definition** (Asymptote). A horizontal, vertical, or oblique asymptote records limiting behavior of a graph near infinity or near a point where the function grows without bound.

## Computational Methods

| Situation                                     | First method to try                 |
|-----------------------------------------------|-------------------------------------|
| Polynomial or rational expression             | Direct substitution, then factoring |
| Radicals                                      | Rationalize with the conjugate      |
| Common factor 0/0                             | Cancel only after factoring         |
| Trigonometric small-angle form                | Use standard trigonometric limits   |
| Bounded expression times vanishing expression | Squeeze theorem                     |
| Large rational functions                      | Compare leading powers              |

*Example* (Factoring a Removable Form). For

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3},$$

factor  $x^2 - 9 = (x - 3)(x + 3)$ , cancel the common nonzero factor for  $x \neq 3$ , and evaluate  $x + 3$  at  $x = 3$ . The limit is 6.

## Theorem Reference

**Theorem** (Limit Laws). *When the relevant limits exist, sums, differences, scalar multiples, products, and quotients of functions have limits computed by applying the same algebraic operation to the limiting values. Quotients require a nonzero denominator limit.*

**Theorem** (Squeeze Theorem). *If  $g(x) \leq f(x) \leq h(x)$  near  $a$  and*

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L,$$

*then  $\lim_{x \rightarrow a} f(x) = L$ .*

## Limits Exercises

*Status: stubbed.*

Exercise sets for limits will be routed here.

## 3.2 Continuity

### 3.2.1 Continuity

#### Definitions

**Definition** (Continuity at a Point). A function  $f$  is continuous at  $a$  when  $f(a)$  is defined,  $\lim_{x \rightarrow a} f(x)$  exists, and

$$\lim_{x \rightarrow a} f(x) = f(a).$$

**Definition** (Continuity on an Interval). A function is continuous on an interval when it is continuous at each interior point and has the appropriate one-sided continuity at any endpoint included in the interval.

**Definition** (Common Discontinuities). A removable discontinuity is a hole, a jump discontinuity has unequal one-sided limiting values, and an infinite discontinuity occurs when function values become unbounded near the point.

## Computational Methods

| Task                     | Practical check                               |
|--------------------------|-----------------------------------------------|
| Polynomial continuity    | Continuous everywhere                         |
| Rational continuity      | Exclude zeros of the denominator              |
| Piecewise continuity     | Check each piece and the junction values      |
| Root/log continuity      | Check the natural domain                      |
| Classify a discontinuity | Compare one-sided behavior and function value |

*Example* (Piecewise Junction). For a piecewise function at  $x = a$ , compute  $f(a)$ ,  $\lim_{x \rightarrow a^-} f(x)$  and  $\lim_{x \rightarrow a^+} f(x)$ . Continuity requires all three quantities to exist and agree.

## Theorem Reference

**Theorem** (Algebra of Continuous Functions). *Sums, differences, scalar multiples, products, quotients with nonzero denominator, and compositions of continuous functions are continuous wherever the resulting expressions are defined.*

**Theorem** (Intermediate Value Theorem). *If  $f$  is continuous on  $[a, b]$  and  $N$  lies between  $f(a)$  and  $f(b)$ , then there is some  $c \in [a, b]$  such that  $f(c) = N$ .*

## Continuity Exercises

*Status: stubbed.*

Exercise sets for continuity will be routed here.

## 3.3 Derivatives

### 3.3.1 Derivatives

#### Definition

**Definition** (Derivative as a Difference Quotient Limit). The derivative of  $f$  at  $a$  is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

when this limit exists.

*Remark* (Slope and Rate of Change). Geometrically,  $f'(a)$  is the slope of the tangent line to  $y = f(x)$  at  $x = a$ . In applications, it is the instantaneous rate of change of the output with respect to the input.

*Example* (Power from the Definition). For  $f(x) = x^2$ ,

$$\frac{f(a+h) - f(a)}{h} = \frac{(a+h)^2 - a^2}{h} = 2a + h,$$

so  $f'(a) = 2a$ . This is a short computational derivation, not a theorem proof vault entry.

## Derivative Algebra

**Theorem** (Constant, Sum, Difference, and Scalar Multiple Rules). *For differentiable functions  $f$  and  $g$  and a constant  $c$ ,*

$$(c)' = 0, \quad (f + g)' = f' + g', \quad (f - g)' = f' - g', \quad (cf)' = cf'.$$

**Theorem** (Product Rule). *If  $f$  and  $g$  are differentiable at  $x$ , then*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

**Theorem** (Quotient Rule). *If  $f$  and  $g$  are differentiable at  $x$  and  $g(x) \neq 0$ , then*

$$\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2}.$$

**Theorem** (Chain Rule). *If  $g$  is differentiable at  $x$  and  $f$  is differentiable at  $g(x)$ , then*

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

## Standard Derivative Table

| Function    | Derivative       |
|-------------|------------------|
| $x^n$       | $nx^{n-1}$       |
| $\sqrt{x}$  | $1/(2\sqrt{x})$  |
| $1/x$       | $-1/x^2$         |
| $e^x$       | $e^x$            |
| $a^x$       | $a^x \ln a$      |
| $\ln x$     | $1/x$            |
| $\log_a x$  | $1/(x \ln a)$    |
| $\sin x$    | $\cos x$         |
| $\cos x$    | $-\sin x$        |
| $\tan x$    | $\sec^2 x$       |
| $\arcsin x$ | $1/\sqrt{1-x^2}$ |
| $\arctan x$ | $1/(1+x^2)$      |

## Chain-Rule Patterns

| Outer form    | Pattern                |
|---------------|------------------------|
| $(u(x))^n$    | $n(u(x))^{n-1}u'(x)$   |
| $e^{u(x)}$    | $e^{u(x)}u'(x)$        |
| $\ln(u(x))$   | $u'(x)/u(x)$           |
| $\sin(u(x))$  | $\cos(u(x))u'(x)$      |
| $\sqrt{u(x)}$ | $u'(x)/(2\sqrt{u(x)})$ |

*Remark* (Method). When differentiating a composition, name the inside expression  $u$ , apply the derivative rule to the outside function, and multiply by  $u'$ .

## Implicit Differentiation

*Remark* (Method). Differentiate both sides with respect to  $x$ , treating  $y$  as a function of  $x$ . Every derivative of an expression involving  $y$  picks up a factor of  $dy/dx$ . Then solve algebraically for  $dy/dx$ .

*Example* (Circle). From  $x^2 + y^2 = 1$ , differentiate to get

$$2x + 2y \frac{dy}{dx} = 0, \quad \frac{dy}{dx} = -\frac{x}{y}.$$

## Logarithmic Differentiation

*Remark* (Method). Use logarithmic differentiation when products, quotients, powers with variable exponents, or many repeated factors make direct differentiation unwieldy. Take logarithms, expand using log laws, differentiate implicitly, and multiply by the original function.

*Example* (Variable Power). If  $y = x^x$  for  $x > 0$ , then

$$\ln y = x \ln x, \quad \frac{y'}{y} = \ln x + 1, \quad y' = x^x(\ln x + 1).$$

## Applications

| Application          | Derivative role                               |
|----------------------|-----------------------------------------------|
| Tangent lines        | Slope at a point                              |
| Linear approximation | $f(a) + f'(a)(x - a)$                         |
| Critical points      | Solve $f'(x) = 0$ or undefined                |
| Curve sketching      | Use $f'$ and $f''$ sign information           |
| Optimization         | Find and compare candidates                   |
| Related rates        | Differentiate a relation with respect to time |

**Theorem** (Mean Value Theorem). If  $f$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ , then there is  $c \in (a, b)$  such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

**Theorem** (L'Hopital's Rule). *For suitable differentiable functions producing an indeterminate form  $0/0$  or  $\infty/\infty$ ,*

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

*when the derivative quotient limit exists under the hypotheses of the rule.*

Exercise-vault records are organized by mutable topic paths. Stable exercise set and item identifiers are canonical and are recorded in `exercise-ledger.yaml`.

## Derivatives

*Status: active.*

### Derivative Rules

**Exercise Set: Derivative Rules** **Stable set ID:** set-derivative-rules-001. **Mode:** computation. **Topic:** derivatives. **Status:** memorialized.

1. **Stable item ID:** ex-deriv-rules-001.

**Exercise.** Find the derivative of

$$y = 7x^5 - 4x^3 + 9x - 2.$$

**Solution.** By the power rule and linearity,

$$\frac{dy}{dx} = 35x^4 - 12x^2 + 9.$$

**Verification.** The result is correct.

2. **Stable item ID:** ex-deriv-rules-002.

**Exercise.** Find the derivative of

$$f(x) = x^2 \sin x.$$

**Solution.** By the product rule,

$$f'(x) = 2x \sin x + x^2 \cos x.$$

**Verification.** The result is corrected. The problem was initially copied as a quotient, but the memorialized exercise is the product  $x^2 \sin x$ .

**Rework note.** Preserve the correction in the ledger so later refactors do not recover the discarded quotient transcription.

### 3. Stable item ID: ex-deriv-rules-003.

**Exercise.** Find the derivative of

$$y = \frac{x^2 + 1}{x - 3}.$$

**Solution.** By the quotient rule,

$$\frac{dy}{dx} = \frac{(x - 3)(2x) - (x^2 + 1)}{(x - 3)^2} = \frac{x^2 - 6x - 1}{(x - 3)^2}.$$

**Verification.** The result is correct, with the final simplification added.

## Mixed Derivative Practice

**Exercise Set: Mixed Derivative Practice**   **Stable set ID: set-derivative-mixed-001.**   **Mode:** computation.   **Topic:** derivatives.   **Status:** generated placeholder.

*This exercise set is reserved for future mixed derivative practice. No solved exercises have been memorialized here yet.*

## 3.4 Integrals

### 3.4.1 Integrals

#### Definitions

**Definition** (Antiderivative). An antiderivative of  $f$  on an interval is a function  $F$  such that  $F' = f$  on that interval.

**Definition** (Definite Integral). The definite integral  $\int_a^b f(x) dx$  measures signed accumulation of  $f$  over  $[a, b]$ . Computationally, it is evaluated by antiderivatives whenever the Fundamental Theorem of Calculus applies.

**Theorem** (Fundamental Theorem of Calculus). If  $F' = f$  on an interval containing  $a$  and  $b$ , then

$$\int_a^b f(x) dx = F(b) - F(a).$$

#### Integral Algebra

| Rule                | Formula                                |
|---------------------|----------------------------------------|
| Linearity           | $\int (af + bg) = a \int f + b \int g$ |
| Interval additivity | $\int_a^b f + \int_b^c f = \int_a^c f$ |
| Reversal            | $\int_a^b f = - \int_b^a f$            |
| Zero width          | $\int_a^a f = 0$                       |
| Constant multiple   | $\int_a^b c dx = c(b - a)$             |

## Techniques

| Technique                  | Use when                                                  |
|----------------------------|-----------------------------------------------------------|
| Substitution               | The integrand contains a composition and its derivative   |
| Integration by parts       | The integrand is a product with a simpler derivative      |
| Trigonometric integrals    | Powers of trig functions can be reduced                   |
| Trigonometric substitution | Radicals match $a^2 - x^2$ , $a^2 + x^2$ , or $x^2 - a^2$ |
| Partial fractions          | A rational function has a factorable denominator          |
| Improper integrals         | The interval is infinite or the integrand is unbounded    |

*Remark* (Method). Before choosing a technique, simplify algebraically, check for a direct antiderivative, and look for an inside function whose derivative is also present.

## Applications

| Application      | Typical integral                             |
|------------------|----------------------------------------------|
| Area             | $\int_a^b (\text{top} - \text{bottom}) dx$   |
| Volume by slices | $\int_a^b A(x) dx$                           |
| Volume by shells | $\int 2\pi(\text{radius})(\text{height}) dx$ |
| Arc length       | $\int_a^b \sqrt{1 + (y')^2} dx$              |
| Surface area     | $\int 2\pi(\text{radius}) ds$                |

## Integrals Exercises

*Status: stubbed.*

Exercise sets for integrals will be routed here.

## 3.5 Sequences and Series

### 3.5.1 Sequences and Series

#### Sequences

**Definition** (Sequence). A sequence is a function from positive integers to numbers, usually written  $(a_n)$ .

**Definition** (Convergence of a Sequence). The sequence  $(a_n)$  converges to  $L$  when  $a_n$  approaches  $L$  as  $n \rightarrow \infty$ . Computationally, this is evaluated using algebraic simplification, standard limits, squeezing, monotonicity, and comparison.

## Series

**Definition** (Series and Partial Sums). The series  $\sum_{n=1}^{\infty} a_n$  is studied through its partial sums

$$s_N = \sum_{n=1}^N a_n.$$

The series converges when the sequence  $(s_N)$  converges.

| Series                   | Convergence pattern     |
|--------------------------|-------------------------|
| Geometric $\sum ar^n$    | Converges for $ r  < 1$ |
| $p$ -series $\sum 1/n^p$ | Converges for $p > 1$   |

## Convergence Tests

| Test                    | Best used when                                     |
|-------------------------|----------------------------------------------------|
| Term test               | Terms do not tend to zero                          |
| Comparison              | Positive terms resemble a known benchmark          |
| Limit comparison        | Ratios to a benchmark have a finite positive limit |
| Integral test           | Terms come from a positive decreasing function     |
| Alternating series test | Signs alternate and magnitudes decrease to zero    |
| Ratio test              | Factorials or exponentials appear                  |
| Root test               | $n$ th powers appear                               |

## Power Series

**Definition** (Power Series). A power series centered at  $a$  has the form

$$\sum_{n=0}^{\infty} c_n (x - a)^n.$$

Its convergence is organized by its radius and interval of convergence.

**Definition** (Taylor and Maclaurin Series). The Taylor series of  $f$  centered at  $a$  is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n.$$

The Maclaurin series is the special case  $a = 0$ .

## Sequences and Series Exercises

*Status: stubbed.*

Exercise sets for sequences and series will be routed here.

## 3.6 Multivariable Calculus

### 3.6.1 Multivariable Calculus

#### Vectors and Curves

**Definition** (Vectors in  $\mathbb{R}^2$  and  $\mathbb{R}^3$ ). Vectors encode magnitude and direction. In  $\mathbb{R}^2$  and  $\mathbb{R}^3$ , they are manipulated through component arithmetic, dot products, cross products, projections, and parametrized lines and planes.

**Definition** (Vector-Valued Function). A vector-valued function  $\mathbf{r}(t)$  traces a curve. Its derivative  $\mathbf{r}'(t)$  is the velocity vector and points in the tangent direction when nonzero.

#### Partial Derivatives

**Definition** (Partial Derivative). For a function  $f(x, y)$ , the partial derivative  $f_x$  differentiates with respect to  $x$  while holding  $y$  constant. The derivative  $f_y$  does the same with respect to  $y$ .

*Remark* (Several-Variable Chain Rule). The chain rule in several variables tracks every path by which a variable can affect the output. In computations, draw the dependency diagram before writing the derivative formula.

**Definition** (Tangent Plane). For  $z = f(x, y)$ , the tangent plane at  $(a, b)$  is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b),$$

when the required differentiability hypotheses hold.

#### Gradients and Directional Derivatives

**Definition** (Gradient). The gradient of  $f(x, y, z)$  is

$$\nabla f = \langle f_x, f_y, f_z \rangle.$$

It points in the direction of steepest increase.

**Definition** (Directional Derivative). For a unit vector  $\mathbf{u}$ ,

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}.$$

*Remark* (Lagrange Multipliers). To optimize  $f$  subject to a constraint  $g = c$ , solve

$$\nabla f = \lambda \nabla g, \quad g = c,$$

then compare candidate values.

## Multiple Integrals

**Definition** (Double and Triple Integrals). Double and triple integrals accumulate a quantity over a region in the plane or in space:

$$\iint_R f \, dA, \quad \iiint_E f \, dV.$$

| Coordinate system | Common use                       |
|-------------------|----------------------------------|
| Polar             | Circular regions in the plane    |
| Cylindrical       | Circular symmetry around an axis |
| Spherical         | Radial symmetry around a point   |

## Vector Calculus Theorems

**Definition** (Line and Surface Integrals). Line integrals accumulate along curves. Surface integrals accumulate over surfaces, often measuring flux through the surface.

**Theorem** (Green's Theorem). *For suitable planar regions and vector fields, a circulation integral around the boundary equals a double integral of curl over the region.*

**Theorem** (Stokes' Theorem). *For suitable oriented surfaces, a circulation integral around the boundary equals a surface integral of curl over the surface.*

**Theorem** (Divergence Theorem). *For suitable solid regions, the flux across the boundary equals the triple integral of divergence over the region.*

## Multivariable Calculus Exercises

*Status: stubbed.*

Exercise sets for multivariable calculus will be routed here.

## Proofs

# Chapter 4

## Geometric Modeling



### 4.1 Curves and Surfaces

#### Parametric Curves and Surfaces — Quick Reference

|                            |                                                                                                     |
|----------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Curve</b>               | $C(u) : [a, b] \rightarrow \mathbb{R}^n$ ; vector-valued, one parameter.                            |
| <b>Velocity</b>            | $C'(u)$ ; tangent direction and speed of traversal.                                                 |
| <b>Unit tangent</b>        | $\hat{T}(u) = C'(u)/\ C'(u)\ $ at a regular point.                                                  |
| <b>Surface</b>             | $S(u, v) : R \rightarrow \mathbb{R}^3$ ; vector-valued, two parameters.                             |
| <b>Partial derivatives</b> | $S_u = \partial S / \partial u$ , $S_v = \partial S / \partial v$ ; tangent to isoparametric lines. |
| <b>Unit normal</b>         | $N = (S_u \times S_v) / \ S_u \times S_v\ $ at a regular point.                                     |
| <b>Tangent plane</b>       | Spanned by $S_u$ and $S_v$ at the base point.                                                       |
| <b>Directional deriv.</b>  | $D_{\mathbf{v}}S = \nabla S \cdot \hat{\mathbf{v}}$ ; rate of change in direction $\mathbf{v}$ .    |

## 4.1.1 Parametric Curves and Surfaces

### Parametric Curves

#### Definition (Parametric Curve)

A *parametric curve* in  $\mathbb{R}^n$  is a vector-valued function

$$C : [a, b] \rightarrow \mathbb{R}^n, \quad C(u) = (x_1(u), x_2(u), \dots, x_n(u)),$$

where each coordinate function  $x_i(u)$  is a real-valued function of the scalar parameter  $u$ . The interval  $[a, b]$  is typically normalised to  $[0, 1]$ .

#### Definition (Velocity, Regularity, Unit Tangent)

The *velocity vector* (or *tangent vector*) at  $u$  is the derivative

$$C'(u) = (x'_1(u), x'_2(u), \dots, x'_n(u)).$$

The curve is *regular* at  $u$  if  $C'(u) \neq \mathbf{0}$ . At a regular point, the *unit tangent vector* is

$$\hat{T}(u) = \frac{C'(u)}{\|C'(u)\|}.$$

The scalar  $\|C'(u)\|$  is the *speed* of traversal.

*Remark* (Properties of parametric curves). Parametric curves carry structure that implicit curves do not:

- **Direction.** The parameter increases from  $a$  to  $b$ , giving the curve a natural orientation. Reversing the parametrisation reverses the direction.
- **Velocity and speed.**  $C'(u)$  measures both the direction of motion and how fast the curve is being traversed. A curve can be *reparametrised* (change of variable  $u = \phi(t)$ ) without changing its shape; the velocity changes but the trace does not.
- **Endpoint interpolation.** The start and end points are  $C(a)$  and  $C(b)$ , accessible directly.
- **Boundedness.** The curve is defined on a closed interval, so it represents a bounded segment, not an infinite line.
- **Cusps.** Where  $C'(u) = \mathbf{0}$ , the curve may have a cusp (velocity drops to zero, direction may reverse). These are degenerate points excluded by the regularity condition.

## Parametric Surfaces

### Definition (Parametric Surface)

A *parametric surface* in  $\mathbb{R}^3$  is a vector-valued function of two parameters,

$$S : R \longrightarrow \mathbb{R}^3, \quad S(u, v) = (x(u, v), y(u, v), z(u, v)),$$

where  $R \subseteq \mathbb{R}^2$  is the *parameter domain* (typically the unit square  $[0, 1]^2$ ).

## Partial Derivatives and Isoparametric Curves

### Definition (Partial Derivative Vectors)

The *partial derivative vectors* of  $S$  at  $(u, v)$  are

$$S_u(u, v) = \frac{\partial S}{\partial u} = \left( \frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right), \quad S_v(u, v) = \frac{\partial S}{\partial v} = \left( \frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right).$$

$S_u$  is the tangent vector to the surface in the  $u$ -direction: it is the velocity of the curve obtained by fixing  $v$  and varying  $u$ .  $S_v$  is the corresponding tangent vector in the  $v$ -direction.

### Definition (Isoparametric Curves)

Fixing one parameter on  $S(u, v)$  produces a curve on the surface:

- $u$ -curve at  $v = v_0$ :  $C_{v_0}(u) = S(u, v_0)$ , tangent =  $S_u(u, v_0)$ .
- $v$ -curve at  $u = u_0$ :  $C_{u_0}(v) = S(u_0, v)$ , tangent =  $S_v(u_0, v)$ .

These *isoparametric curves* (isocurves) sweep the surface and form its parameter grid. The two families intersect at every surface point  $S(u_0, v_0)$ .

## Directional Derivative

### Definition (Directional Derivative on a Surface)

The *directional derivative* of  $S$  at  $(u_0, v_0)$  in the direction  $\mathbf{d} = (d_u, d_v)$  (a unit vector in parameter space) is

$$D_{\mathbf{d}} S = d_u S_u(u_0, v_0) + d_v S_v(u_0, v_0).$$

It measures the rate of change of position on the surface as the parameter moves in direction  $\mathbf{d}$  in the  $uv$ -plane. Every tangent vector to the surface at  $(u_0, v_0)$  is a directional derivative for some choice of  $\mathbf{d}$ .

## Unit Normal and Tangent Plane

### Definition (Regularity, Unit Normal, Tangent Plane)

The surface is *regular* at  $(u_0, v_0)$  if  $S_u \times S_v \neq \mathbf{0}$  there.

At a regular point, the *unit normal vector* is

$$N(u_0, v_0) = \frac{S_u(u_0, v_0) \times S_v(u_0, v_0)}{\|S_u(u_0, v_0) \times S_v(u_0, v_0)\|}.$$

The *tangent plane* at  $S(u_0, v_0)$  is the plane through  $S(u_0, v_0)$  spanned by  $S_u$  and  $S_v$ :

$$\Pi = \{ S(u_0, v_0) + \alpha S_u + \beta S_v \mid \alpha, \beta \in \mathbb{R} \}.$$

Its equation in terms of position  $\mathbf{p}$  is

$$N \cdot (\mathbf{p} - S(u_0, v_0)) = 0.$$

*Remark* (Geometric meaning of  $S_u \times S_v$ ). The magnitude  $\|S_u \times S_v\|$  equals the area of the parallelogram spanned by  $S_u$  and  $S_v$ , which is the *local area element* of the surface. Where  $\|S_u \times S_v\| = 0$  the surface is degenerate (the two tangent directions are parallel or one vanishes), and the normal is not defined.

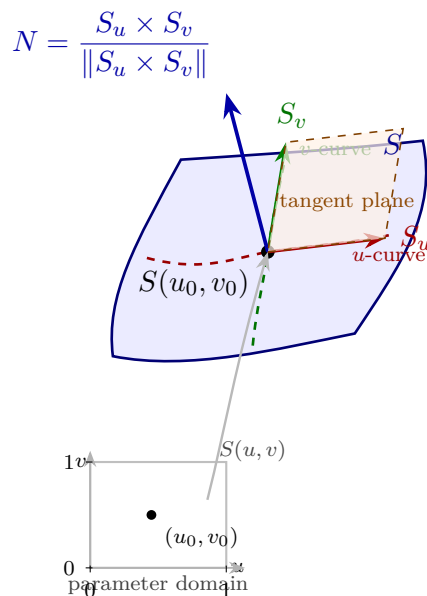


Figure: A parametric surface patch  $S(u, v)$  with its parameter domain (lower left). At the point  $S(u_0, v_0)$ , the partial derivative vectors  $S_u$  (red) and  $S_v$  (green) are tangent to the isoparametric curves; together they span the tangent plane (orange). The unit normal  $N = S_u \times S_v / \|S_u \times S_v\|$  (blue) is perpendicular to the tangent plane.

Status: Active

This section is actively expanding alongside work in Piegl & Tiller, *The NURBS Book*. Subsequent sections on Bézier curves, B-splines, NURBS, and continuity conditions follow.

## 4.2 Bezier Curves

### Proofs

# Chapter 5

## Computational Geometry



### 5.1 Convex Hulls

#### Toolkit: Convex Hulls

|                          |                                                                     |
|--------------------------|---------------------------------------------------------------------|
| <b>Ambient space</b>     | Finite point sets in $\mathbb{R}^2$ .                               |
| <b>Convexity</b>         | Line segments and convex combinations.                              |
| <b>Certificates</b>      | Extreme points, supporting lines, and maximizing directions.        |
| <b>Order data</b>        | Distinct $x$ -coordinates and slopes between ordered points.        |
| <b>Algorithmic shape</b> | Build the upper hull by repeatedly choosing the next maximal slope. |

*Remark* (Exposition). Computational geometry studies finite geometric data, such as point sets in the plane, using exact definitions and algorithms. Convex hulls are the first organizing example: they connect geometry, order and maximization, supporting lines, and algorithmic construction.

## 5.2 Convex Hulls

### Convexity and Hulls

#### Convexity and Convex Hulls

This formulation makes precise the geometric idea of filling in all straight segments forced by a point set. Convex combinations give an algebraic language for the same operation, and the convex hull records the smallest set closed under that operation.

#### Definition

**Definition 5.2.1** (Convex Set). A set  $S \subseteq \mathbb{R}^2$  is *convex* if for every pair of points  $a, b \in S$  and every  $\lambda \in [0, 1]$ ,

$$\lambda a + (1 - \lambda)b \in S.$$

Equivalently,  $S$  contains the line segment between any two of its points.

*Remark* (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

#### Definition

**Definition 5.2.2** (Convex Combination). A *convex combination* of points  $q_1, \dots, q_n \in \mathbb{R}^2$  is a point of the form

$$\sum_{i=1}^n \lambda_i q_i, \quad \lambda_i \geq 0 \text{ for all } i, \quad \sum_{i=1}^n \lambda_i = 1.$$

The numbers  $\lambda_i$  are called the weights.

*Remark* (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

#### Definition

**Definition 5.2.3** (Convex Hull). The *convex hull* of a set  $P \subseteq \mathbb{R}^2$ , written  $\text{conv}(P)$ , is the set of all convex combinations of finitely many points of  $P$ :

$$\text{conv}(P) = \left\{ \sum_{i=1}^n \lambda_i q_i \mid n \geq 1, q_i \in P, \lambda_i \geq 0, \sum_{i=1}^n \lambda_i = 1 \right\}.$$

Equivalently,  $\text{conv}(P)$  is the smallest convex set containing  $P$ , i.e. the intersection of all convex sets that contain  $P$ .

*Remark* (Standard quantified statement). The statement applies to the objects named in the

hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

#### Lemma

**Lemma 5.2.4** (Convex hull as the smallest convex set). *Let  $P \subseteq \mathbb{R}^2$ . Then  $\text{conv}(P)$  is convex, contains  $P$ , and is contained in every convex set that contains  $P$ . Hence*

$$\text{conv}(P) = \bigcap \{K \subseteq \mathbb{R}^2 : K \text{ is convex and } P \subseteq K\}.$$

*Remark* (Proof). See [the proof stub](#).

*Remark* (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

*Remark* (Dependencies).

- [Convex Set](#)
- [Convex Combination](#)
- [Convex Hull](#)

## Supporting Functionals

### Supporting Functionals and Vertices

This topic isolates the passage from geometry to optimization. A supporting line is a geometric certificate that a convex set lies on one side of a boundary, while a linear functional turns that certificate into a maximum.

#### Definition

**Definition 5.2.5** (Extreme Point). Let  $S \subseteq \mathbb{R}^2$  be convex. A point  $p \in S$  is an *extreme point* of  $S$  if whenever

$$p = \lambda a + (1 - \lambda)b, \quad a, b \in S, \quad \lambda \in (0, 1),$$

it follows that  $a = b = p$ . For a finite point set  $P$ , the extreme points of  $\text{conv}(P)$  are called the vertices of the convex hull.

*Remark* (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

### Definition

**Definition 5.2.6** (Supporting Line). A line  $\ell \subseteq \mathbb{R}^2$  is a *supporting line* of a convex set  $S$  at a point  $p \in S$  if  $p \in \ell$  and  $S$  lies entirely in one of the two closed half-planes bounded by  $\ell$ . Equivalently, there exists  $d \in \mathbb{R}^2 \setminus \{0\}$  such that

$$\ell = \{x \in \mathbb{R}^2 : \langle d, x \rangle = \langle d, p \rangle\}$$

and

$$\langle d, q \rangle \leq \langle d, p \rangle \quad \text{for all } q \in S.$$

*Remark* (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

### Lemma

**Lemma 5.2.7** (Linear functionals preserve convex combinations). *Let  $d \in \mathbb{R}^2$ , and let*

$$z = \sum_{i=1}^n \lambda_i q_i$$

*be a convex combination of points  $q_1, \dots, q_n \in \mathbb{R}^2$ . Then*

$$\langle d, z \rangle = \sum_{i=1}^n \lambda_i \langle d, q_i \rangle.$$

*Consequently, if  $\langle d, q_i \rangle \leq M$  for every  $i$ , then*

$$\langle d, z \rangle \leq M.$$

*Remark* (Proof). See [the proof stub](#).

*Remark* (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

### Theorem

*Remark* (Dependencies).

- Convex Combination
- Addition on  $\mathbb{R}$
- Multiplication on  $\mathbb{R}$

**Theorem 5.2.8** (Supporting functional characterization of hull vertices). *Let  $P \subset \mathbb{R}^2$  be finite and in general position. A point  $p \in P$  is a vertex of  $\text{conv}(P)$  if and only if there exists a direction  $d \in \mathbb{R}^2 \setminus \{0\}$  such that*

$$\langle d, p \rangle > \langle d, q \rangle \quad \text{for all } q \in P \setminus \{p\}.$$

*Equivalently,  $p$  is the unique point of  $P$  attaining*

$$\max_{q \in P} \langle d, q \rangle.$$

*Thus every hull vertex is a maximum of  $P$  under some linear functional.*

*Remark* (Proof). See [the proof stub](#).

*Remark* (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

*Remark* (Finite maximum form). The maximum is used here because  $P$  is finite. This is the finite-set version of the same supporting-functional intuition that appears in supremum language for more general sets.

*Remark* (Dependencies).

- Convex Combination
- Convex Hull
- Extreme Point
- Supporting Line
- Linear Functionals Preserve Convex Combinations

## Upper Hull Structure

### Monotone Slopes and Greedy Construction

This topic converts the supporting-line description into an ordered planar construction. Under the general-position assumptions, each relevant maximum is unique, and the upper hull can be recognized by the strict monotonicity of successive slopes.

**Assumption 5.2.9** (General Position for the Hull Theorems). Let  $P \subset \mathbb{R}^2$  be a finite set in general position: the points have distinct  $x$ -coordinates and no three points are collinear. For  $p \in \mathbb{R}^2$ , write  $x(p)$  and  $y(p)$  for its coordinates.

#### Theorem

**Theorem 5.2.10** (Monotone slope characterization of the upper hull). *Let  $P \subset \mathbb{R}^2$  be finite and in general position. Let*

$$v_0, v_1, \dots, v_m$$

*be the vertices of the upper hull of  $P$ , listed in increasing  $x$ -coordinate, and define*

$$s_k = \frac{y(v_{k+1}) - y(v_k)}{x(v_{k+1}) - x(v_k)}, \quad k = 0, 1, \dots, m-1.$$

*Then*

$$s_0 > s_1 > \dots > s_{m-1}.$$

*Conversely, if points  $w_0, \dots, w_m$  satisfy*

$$x(w_0) < x(w_1) < \dots < x(w_m)$$

*and their consecutive slopes are strictly decreasing, then every  $w_k$  is a vertex of*

$$\text{conv}(\{w_0, \dots, w_m\}).$$

*The lower hull is characterized similarly, with strictly increasing consecutive slopes.*

*Remark* (Proof). See [the proof stub](#).

*Remark* (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

## Theorem

*Remark* (Dependencies).

- Convex Hull
- Supporting Line
- Linear Functionals Preserve Convex Combinations
- Supporting Functional Characterization of Hull Vertices

**Theorem 5.2.11** (Greedy slope construction of the upper hull). *Let  $P \subset \mathbb{R}^2$  be finite and in general position. Define*

$$v_0 = \arg \min_{p \in P} x(p).$$

*Given  $v_k$  with*

$$x(v_k) < \max_{p \in P} x(p),$$

*define*

$$v_{k+1} = \arg \max_{\substack{q \in P \\ x(q) > x(v_k)}} \frac{y(q) - y(v_k)}{x(q) - x(v_k)}.$$

*Terminate once*

$$v_k = \arg \max_{p \in P} x(p).$$

*Then the points*

$$v_0, v_1, \dots, v_m$$

*produced by this construction are exactly the vertices of the upper hull of  $P$ , and the slopes encountered along the way are strictly decreasing.*

*Remark* (Proof). See [the proof stub](#).

*Remark* (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

*Remark* (Uniqueness of choices). The distinct  $x$ -coordinates make the initial and terminal  $x$ -extreme points unique. The no-three-collinear condition makes the slope maximizer in the greedy step unique.

*Remark* (Proof roadmap). The proof ladder is:

convex combinations  $\rightarrow$  linear functionals  $\rightarrow$  supporting lines  $\rightarrow$  monotone slopes  $\rightarrow$  greedy hull construction

The first two lemmas are algebraic. The monotone slope theorem is an analytic-geometry

statement about three ordered points. The greedy construction theorem is the algorithmic consequence.

*Remark* (Dependencies).

- Convex Hull
- Supporting Line
- Linear Functionals Preserve Convex Combinations
- Supporting Functional Characterization of Hull Vertices
- Monotone slope characterization of the upper hull

## Proofs

*Remark* (Return). [Return to Theorem](#)

*Remark* (Proof vault). [Handwritten proof record](#)

**Lemma** (Convex hull as the smallest convex set). *Let  $P \subseteq \mathbb{R}^2$ . Then  $\text{conv}(P)$  is convex, contains  $P$ , and is contained in every convex set that contains  $P$ . Hence*

$$\text{conv}(P) = \bigcap \{K \subseteq \mathbb{R}^2 : K \text{ is convex and } P \subseteq K\}.$$

*Professional Standard Proof.*

TODO: professional standard proof for computational-geometry-convex-hull-smallest-convex-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for computational-geometry-convex-hull-smallest-convex-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Convex Set](#)
- [Convex Combination](#)
- [Convex Hull](#)

*Remark* (Return). [Return to Theorem](#)

*Remark* (Proof vault). [Handwritten proof record](#)

**Lemma** (Linear functionals preserve convex combinations). *Let  $d \in \mathbb{R}^2$ , and let*

$$z = \sum_{i=1}^n \lambda_i q_i$$

*be a convex combination of points  $q_1, \dots, q_n \in \mathbb{R}^2$ . Then*

$$\langle d, z \rangle = \sum_{i=1}^n \lambda_i \langle d, q_i \rangle.$$

*Consequently, if  $\langle d, q_i \rangle \leq M$  for every  $i$ , then*

$$\langle d, z \rangle \leq M.$$

*Professional Standard Proof.*

TODO: professional standard proof for computational-geometry-linear-functionals-convex-combinations. □

*Detailed Learning Proof.*

TODO: detailed learning proof for computational-geometry-linear-functionals-convex-combinations ■  
□

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

*Remark* (Proof vault). [Handwritten proof record](#)

**Theorem** (Supporting functional characterization of hull vertices). *Let  $P \subset \mathbb{R}^2$  be finite and in general position. A point  $p \in P$  is a vertex of  $\text{conv}(P)$  if and only if there exists a direction  $d \in \mathbb{R}^2 \setminus \{0\}$  such that*

$$\langle d, p \rangle > \langle d, q \rangle \quad \text{for all } q \in P \setminus \{p\}.$$

*Equivalently,  $p$  is the unique point of  $P$  attaining*

$$\max_{q \in P} \langle d, q \rangle.$$

*Thus every hull vertex is a maximum of  $P$  under some linear functional.*

*Professional Standard Proof.*

TODO: Prove one direction by using a unique maximizing functional to create a supporting line at  $p$ . Prove the converse by producing a supporting functional for a hull vertex under the finite general-position hypotheses.  $\square$

*Detailed Learning Proof.*

TODO: Expand the two directions, making explicit where finiteness, supporting lines, and uniqueness from general position enter.  $\square$

*Remark* (Proof structure). The proof translates between extreme points, supporting lines, and strict maximizers of linear functionals.

*Remark* (Dependencies).

- [Convex Hull](#)
- [Extreme Point](#)
- [Supporting Line](#)
- [Linear Functionals Preserve Convex Combinations](#)
- TODO: finite set maximum

*Remark* (Return). [Return to Theorem](#)

*Remark* (Proof vault). [Handwritten proof record](#)

**Theorem** (Monotone slope characterization of the upper hull). *Let  $P \subset \mathbb{R}^2$  be finite and in general position. Let*

$$v_0, v_1, \dots, v_m$$

*be the vertices of the upper hull of  $P$ , listed in increasing  $x$ -coordinate, and define*

$$s_k = \frac{y(v_{k+1}) - y(v_k)}{x(v_{k+1}) - x(v_k)}, \quad k = 0, 1, \dots, m-1.$$

*Then*

$$s_0 > s_1 > \dots > s_{m-1}.$$

*Conversely, if points  $w_0, \dots, w_m$  satisfy*

$$x(w_0) < x(w_1) < \dots < x(w_m)$$

*and their consecutive slopes are strictly decreasing, then every  $w_k$  is a vertex of*

$$\text{conv}(\{w_0, \dots, w_m\}).$$

*The lower hull is characterized similarly, with strictly increasing consecutive slopes.*

*Professional Standard Proof.*

TODO: professional standard proof for computational-geometry-monotone-slope-upper-hull. □

*Detailed Learning Proof.*

TODO: detailed learning proof for computational-geometry-monotone-slope-upper-hull. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

*Remark* (Proof vault). [Handwritten proof record](#)

**Theorem** (Greedy slope construction of the upper hull). *Let  $P \subset \mathbb{R}^2$  be finite and in general position. Define*

$$v_0 = \arg \min_{p \in P} x(p).$$

*Given  $v_k$  with*

$$x(v_k) < \max_{p \in P} x(p),$$

*define*

$$v_{k+1} = \arg \max_{\substack{q \in P \\ x(q) > x(v_k)}} \frac{y(q) - y(v_k)}{x(q) - x(v_k)}.$$

*Terminate once*

$$v_k = \arg \max_{p \in P} x(p).$$

*Then the points*

$$v_0, v_1, \dots, v_m$$

*produced by this construction are exactly the vertices of the upper hull of  $P$ , and the slopes encountered along the way are strictly decreasing.*

*Professional Standard Proof.*

TODO: professional standard proof for computational-geometry-greedy-slope-upper-hull. □■

*Detailed Learning Proof.*

TODO: detailed learning proof for computational-geometry-greedy-slope-upper-hull. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Convex Hull](#)
- [Supporting Line](#)
- [Linear Functionals Preserve Convex Combinations](#)
- [Supporting Functional Characterization of Hull Vertices](#)
- TODO: finite set maximum
- TODO: slope of a line

# Chapter 6

## Computational Linear Algebra



### 6.1 Vectors

#### Vectors and Linear Combinations — Quick Reference

|                                            |                                                                                                                                                        |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vector in <math>\mathbb{R}^n</math></b> | An ordered list $(v_1, v_2, \dots, v_n)$ of $n$ real numbers.                                                                                          |
| <b>Addition</b>                            | $(v_1, \dots, v_n) + (w_1, \dots, w_n) = (v_1 + w_1, \dots, v_n + w_n)$ .                                                                              |
| <b>Scalar multiplication</b>               | $c(v_1, \dots, v_n) = (cv_1, \dots, cv_n)$ .                                                                                                           |
| <b>Linear combination</b>                  | $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k$ ; scalars times vectors, summed.                                                         |
| <b>Convex combination</b>                  | Linear combination with $c_i \geq 0$ and $\sum c_i = 1$ ; result lies inside the convex hull.                                                          |
| <b>Standard basis vectors</b>              | $\mathbf{e}_j \in \mathbb{R}^n$ : all zeros except a 1 in position $j$ . Every vector is a linear combination of $\mathbf{e}_1, \dots, \mathbf{e}_n$ . |

## Matrices — Quick Reference

|                      |                                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Matrix</b>        | Rectangular array of numbers; $A \in \mathbb{R}^{m \times n}$ has $m$ rows, $n$ columns.                                   |
| <b>Matrix–vector</b> | $(A\mathbf{x})_i = \sum_j A_{ij}x_j$ ; $A\mathbf{x}$ is a linear combination of columns of $A$ .                           |
| <b>Matrix–matrix</b> | $(AB)_{ij} = \sum_k A_{ik}B_{kj}$ ; requires inner dimensions to match.                                                    |
| <b>Transpose</b>     | $(A^T)_{ij} = A_{ji}$ ; rows become columns.                                                                               |
| <b>Identity</b>      | $I_n$ : ones on diagonal, zeros elsewhere; $IA = AI = A$ .                                                                 |
| <b>Column view</b>   | $A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$ where $\mathbf{a}_j$ are the columns of $A$ . |

## Lengths and Dot Products — Quick Reference

|                       |                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------|
| <b>Dot product</b>    | $\mathbf{v} \cdot \mathbf{w} = v_1w_1 + v_2w_2 + \cdots + v_nw_n \in \mathbb{R}$ .      |
| <b>Length (norm)</b>  | $\ \mathbf{v}\  = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + \cdots + v_n^2}$ . |
| <b>Unit vector</b>    | $\hat{\mathbf{v}} = \mathbf{v}/\ \mathbf{v}\ $ ; has length 1.                          |
| <b>Angle</b>          | $\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\ \mathbf{v}\  \ \mathbf{w}\ }$ .     |
| <b>Orthogonality</b>  | $\mathbf{v} \perp \mathbf{w}$ iff $\mathbf{v} \cdot \mathbf{w} = 0$ .                   |
| <b>Cauchy–Schwarz</b> | $ \mathbf{v} \cdot \mathbf{w}  \leq \ \mathbf{v}\  \ \mathbf{w}\ $ .                    |

### 6.1.1 Vectors and Linear Combinations

#### Vectors

##### Definition (Vector in $\mathbb{R}^n$ )

A *vector* in  $\mathbb{R}^n$  is an ordered list of  $n$  real numbers, written  $\mathbf{v} = (v_1, v_2, \dots, v_n)$ . The number  $n$  is the *dimension* of the vector.

##### Vector addition:

$$\mathbf{v} + \mathbf{w} = (v_1 + w_1, v_2 + w_2, \dots, v_n + w_n).$$

##### Scalar multiplication:

$$c\mathbf{v} = (cv_1, cv_2, \dots, cv_n), \quad c \in \mathbb{R}.$$

*Remark* (Geometric meaning). In  $\mathbb{R}^2$  and  $\mathbb{R}^3$ , a vector is an arrow indicating displacement. Its entries specify length and direction, but *not* location — the same vector can be placed anywhere in space. Addition corresponds to following one displacement and then another (head-to-tail rule). Scalar multiplication stretches or shrinks the arrow; negative scalars reverse its direction.

### Theorem (Properties of Vector Operations)

For all  $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$  and  $c, d \in \mathbb{R}$ :

1.  $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$  (commutativity)
2.  $(\mathbf{v} + \mathbf{w}) + \mathbf{x} = \mathbf{v} + (\mathbf{w} + \mathbf{x})$  (associativity)
3.  $c(\mathbf{v} + \mathbf{w}) = c\mathbf{v} + c\mathbf{w}$  (distributivity over vector addition)
4.  $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$  (distributivity over scalar addition)
5.  $c(d\mathbf{v}) = (cd)\mathbf{v}$  (compatibility)

## Linear Combinations

### Definition (Linear Combination)

A *linear combination* of vectors  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$  is any vector of the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k,$$

where  $c_1, c_2, \dots, c_k \in \mathbb{R}$  are called the *coefficients*.

A *convex combination* is a linear combination in which  $c_i \geq 0$  for all  $i$  and  $\sum_{i=1}^k c_i = 1$ .

*Remark* (Why convex combinations matter for NURBS). Every point on a Bézier or B-spline curve is a convex combination of control points. The Bernstein basis functions  $B_{i,n}(u)$  are non-negative and sum to 1, so  $C(u) = \sum_{i=0}^n B_{i,n}(u)P_i$  is a convex combination of the control points  $\{P_i\}$  at every parameter value  $u$ . This is the *convex hull property*: the curve lies entirely within the convex hull of its control polygon.

### Definition (Standard Basis Vectors)

For  $j = 1, 2, \dots, n$ , the *j-th standard basis vector*  $\mathbf{e}_j \in \mathbb{R}^n$  has a 1 in position  $j$  and 0s everywhere else. Every vector  $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{R}^n$  can be written as

$$\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \dots + v_n\mathbf{e}_n.$$

## 6.1.2 Lengths and Dot Products

### The Dot Product

#### Definition (Dot Product)

The *dot product* (or *inner product*) of  $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$  is

$$\mathbf{v} \cdot \mathbf{w} = v_1w_1 + v_2w_2 + \dots + v_nw_n \in \mathbb{R}.$$

### Theorem (Properties of the Dot Product)

For all  $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$  and  $c \in \mathbb{R}$ :

1.  $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$  (commutativity)
2.  $\mathbf{v} \cdot (\mathbf{w} + \mathbf{x}) = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{x}$  (distributivity)
3.  $\mathbf{v} \cdot (c\mathbf{w}) = c(\mathbf{v} \cdot \mathbf{w})$  (scalar compatibility)
4.  $\mathbf{v} \cdot \mathbf{v} \geq 0$ , with equality iff  $\mathbf{v} = \mathbf{0}$  (positive definiteness)

## Length and Angle

### Definition (Length and Unit Vector)

The *length* (or *norm*) of  $\mathbf{v} \in \mathbb{R}^n$  is

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

A *unit vector* has length 1. The *normalisation* of a nonzero vector  $\mathbf{v}$  is  $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$ , which is the unit vector in the direction of  $\mathbf{v}$ .

### Theorem (Cauchy–Schwarz Inequality)

For all  $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ :

$$|\mathbf{v} \cdot \mathbf{w}| \leq \|\mathbf{v}\| \|\mathbf{w}\|,$$

with equality if and only if  $\mathbf{v}$  and  $\mathbf{w}$  are parallel (one is a scalar multiple of the other).

### Definition (Angle Between Vectors)

The *angle*  $\theta \in [0, \pi]$  between nonzero vectors  $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$  is defined by

$$\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|}.$$

The vectors are *orthogonal* (written  $\mathbf{v} \perp \mathbf{w}$ ) if  $\theta = \pi/2$ , equivalently if  $\mathbf{v} \cdot \mathbf{w} = 0$ .

*Remark* (Dot product and normals in NURBS). The unit normal to a parametric surface  $S(u, v)$  is computed as  $N = S_u \times S_v / \|S_u \times S_v\|$ , which requires both the cross product and normalisation via the norm. The dot product also enters the convex hull argument: showing that a B-spline curve lies on a specific side of a hyperplane reduces to checking the sign of dot products of control points with the hyperplane's normal.

### 6.1.3 Introduction to Matrices

#### Matrices and Matrix–Vector Multiplication

##### Definition (Matrix)

An  $m \times n$  *matrix*  $A$  is a rectangular array of real numbers with  $m$  rows and  $n$  columns:

$$A = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{pmatrix}.$$

The entry in row  $i$ , column  $j$  is denoted  $A_{ij}$  or  $[A]_{ij}$ . We write  $A \in \mathbb{R}^{m \times n}$ .

##### Definition (Matrix–Vector and Matrix–Matrix Multiplication)

If  $A \in \mathbb{R}^{m \times n}$  and  $\mathbf{x} \in \mathbb{R}^n$ , then  $A\mathbf{x} \in \mathbb{R}^m$  is defined by

$$(A\mathbf{x})_i = \sum_{j=1}^n A_{ij}x_j, \quad i = 1, \dots, m.$$

Equivalently,  $A\mathbf{x}$  is the linear combination of the columns of  $A$  with coefficients from  $\mathbf{x}$ :

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

If  $A \in \mathbb{R}^{m \times n}$  and  $B \in \mathbb{R}^{n \times p}$ , their product  $AB \in \mathbb{R}^{m \times p}$  has entries

$$(AB)_{ij} = \sum_{k=1}^n A_{ik}B_{kj}.$$

*Remark* (Matrix multiplication is not commutative). In general  $AB \neq BA$ , even when both products are defined. Order matters: applying transformation  $B$  first, then  $A$ , is written  $AB$ .

## The Transpose

### Definition (Transpose)

The *transpose* of  $A \in \mathbb{R}^{m \times n}$  is the matrix  $A^T \in \mathbb{R}^{n \times m}$  defined by  $(A^T)_{ij} = A_{ji}$ . Rows of  $A$  become columns of  $A^T$ . Key properties:

- $(A^T)^T = A$
- $(AB)^T = B^T A^T$
- $\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T \mathbf{w}$  (dot product as matrix multiplication)

## Matrix Form of a Curve

*Remark* (NURBS connection: matrix form of curves and surfaces). The power basis curve  $C(u) = \sum_{i=0}^n \mathbf{a}_i u^i$  can be written as

$$C(u) = [\mathbf{a}_0 \quad \mathbf{a}_1 \quad \cdots \quad \mathbf{a}_n] \begin{pmatrix} 1 \\ u \\ \vdots \\ u^n \end{pmatrix} = A \mathbf{u}(u),$$

where  $A$  is the matrix of coefficient vectors and  $\mathbf{u}(u)$  is the monomial vector. A Bézier curve has the equivalent matrix form  $C(u) = [u^i]^T M [P_i]$ , where  $M$  is the  $(n+1) \times (n+1)$  change-of-basis matrix converting Bernstein basis to power basis. A tensor product surface takes the form  $S(u, v) = [f_i(u)]^T [b_{i,j}] [g_j(v)]$ , a matrix sandwiched between two basis-function vectors. All of these representations rely on matrix–vector multiplication as defined above.

## 6.2 Solving Linear Equations

### Matrix Operations — Quick Reference

|                          |                                                               |
|--------------------------|---------------------------------------------------------------|
| <b>Associativity</b>     | $A(BC) = (AB)C.$                                              |
| <b>Distributivity</b>    | $A(B + C) = AB + AC; (A + B)C = AC + BC.$                     |
| <b>NOT commutative</b>   | $AB \neq BA$ in general.                                      |
| <b>Identity</b>          | $AI = IA = A.$                                                |
| <b>Transpose product</b> | $(AB)^T = B^T A^T.$                                           |
| <b>Diagonal matrices</b> | $D_1 D_2$ is diagonal; $DA$ scales rows; $AD$ scales columns. |

### Inverse Matrices — Quick Reference

|                                        |                                                                                                                         |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Inverse</b>                         | $A^{-1}$ : the unique matrix with $AA^{-1} = A^{-1}A = I.$                                                              |
| <b>Existence</b>                       | $A^{-1}$ exists iff $A$ is square and $\det(A) \neq 0$ (nonsingular).                                                   |
| <b><math>2 \times 2</math> formula</b> | $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$ |
| <b>Product rule</b>                    | $(AB)^{-1} = B^{-1}A^{-1}$ (order reverses).                                                                            |
| <b>Transpose rule</b>                  | $(A^T)^{-1} = (A^{-1})^T.$                                                                                              |
| <b>Change of basis</b>                 | If $M$ converts basis $B_1$ to $B_2$ , then $M^{-1}$ converts $B_2$ to $B_1$ .                                          |

## 6.2.1 Rules for Matrix Operations

### Algebraic Rules

#### Theorem (Rules for Matrix Multiplication)

When the dimensions are compatible:

1.  $A(BC) = (AB)C$  (associativity)
2.  $A(B + C) = AB + AC$  (left distributivity)
3.  $(A + B)C = AC + BC$  (right distributivity)
4.  $c(AB) = (cA)B = A(cB)$  for any scalar  $c$
5.  $(AB)^T = B^T A^T$  (transpose reverses order)

The following do **not** hold in general:

- $AB \neq BA$  (not commutative)
- $AB = AC \not\Rightarrow B = C$  (no cancellation in general)
- $AB = 0 \not\Rightarrow A = 0$  or  $B = 0$

### Diagonal Matrices

#### Definition (Diagonal Matrix)

A *diagonal matrix*  $D \in \mathbb{R}^{n \times n}$  has  $D_{ij} = 0$  for all  $i \neq j$ . Written  $D = \text{diag}(d_1, d_2, \dots, d_n)$ .  
Key computational facts:

- $DA$  multiplies each *row*  $i$  of  $A$  by  $d_i$ .
- $AD$  multiplies each *column*  $j$  of  $A$  by  $d_j$ .
- $D_1 D_2 = \text{diag}(d_1 e_1, \dots, d_n e_n)$  where  $e_j$  are the diagonal entries of  $D_2$ .

### Block Matrices

*Remark* (Block matrix multiplication). Matrices can be partitioned into blocks, and multiplied block-by-block as if the blocks were scalar entries, provided the block sizes are compatible. If  $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$  and  $B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$ , then

$$AB = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix},$$

provided inner block dimensions match. The standard matrix of a linear transformation  $[T] = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)]$  is a block matrix with column vectors as blocks, and  $[T]\mathbf{v} = v_1T(\mathbf{e}_1) + \cdots + v_nT(\mathbf{e}_n)$  is block matrix–vector multiplication.

## 6.2.2 Inverse Matrices

### Definition and Properties

#### Definition (Invertible Matrix)

A square matrix  $A \in \mathbb{R}^{n \times n}$  is *invertible* (or *nonsingular*) if there exists a matrix  $A^{-1} \in \mathbb{R}^{n \times n}$  such that

$$AA^{-1} = A^{-1}A = I_n.$$

The matrix  $A^{-1}$  is the *inverse* of  $A$  and is unique. A matrix with no inverse is called *singular*.

#### Theorem (Properties of Inverses)

When  $A$  and  $B$  are invertible  $n \times n$  matrices:

1.  $(A^{-1})^{-1} = A$
2.  $(AB)^{-1} = B^{-1}A^{-1}$  (order reverses)
3.  $(A^T)^{-1} = (A^{-1})^T$
4.  $(cA)^{-1} = \frac{1}{c}A^{-1}$  for  $c \neq 0$

### The $2 \times 2$ Case

#### Formula ( $2 \times 2$ Inverse)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad ad - bc \neq 0.$$

The quantity  $ad - bc$  is the determinant. The matrix is singular when  $ad = bc$ .

### Change of Basis via Inverse

*Remark* (NURBS connection: converting between Bernstein and power basis). In Piegls & Tiller Exercise 1.14, the Bernstein basis functions of degree  $n$  can be written as  $[B_{0,n}(u), \dots, B_{n,n}(u)] = [1, u, \dots, u^n]M$ , where  $M$  is an  $(n+1) \times (n+1)$  matrix. This means a Bézier curve has the matrix form

$$C(u) = [u^i]^T M [P_i],$$

where  $[u^i]^T$  is the monomial row vector,  $[P_i]$  is the column of control points, and  $M$  encodes the change from power basis to Bernstein basis.

The inverse  $M^{-1}$  converts the other direction: if you have a curve in power basis form with coefficients  $[\mathbf{a}_i]$ , then the equivalent Bézier control points are  $[P_i] = M^{-1}[\mathbf{a}_i]$ .

This is a direct application of the matrix inverse:  $M$  converts one basis to another, and  $M^{-1}$  inverts the conversion.

## 6.3 Vector Spaces

### Basis and Dimension — Quick Reference

|                             |                                                                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Linearly independent</b> | No vector is a linear combination of the others; only $\mathbf{0} = c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$ has $c_i = 0$ for all $i$ . |
| <b>Span</b>                 | $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ : all linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_k$ .                       |
| <b>Basis</b>                | A linearly independent set that spans the space.                                                                                            |
| <b>Dimension</b>            | Number of vectors in any basis; all bases have the same size.                                                                               |
| <b>Change of basis</b>      | Transition matrix $P$ : columns are the new basis vectors in terms of the old basis.                                                        |

### 6.3.1 Independence, Basis, and Dimension

#### Linear Independence

##### Definition (Linear Independence)

Vectors  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$  are *linearly independent* if the only solution to

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

is  $c_1 = c_2 = \cdots = c_k = 0$ . Otherwise they are *linearly dependent*: at least one vector is a linear combination of the others.

#### Basis and Dimension

##### Definition (Basis)

A *basis* for a subspace  $V \subseteq \mathbb{R}^n$  is a set of vectors  $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$  that is:

1. linearly independent, and
2. spans  $V$  (every vector in  $V$  is a linear combination of  $\mathbf{v}_1, \dots, \mathbf{v}_k$ ).

##### Theorem (Dimension is Well-Defined)

Any two bases for the same subspace  $V$  have the same number of vectors. This common number is called the *dimension* of  $V$ , written  $\dim V$ .

*Remark.* The standard basis  $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$  is a basis for  $\mathbb{R}^n$ , so  $\dim \mathbb{R}^n = n$ . The key insight: every set of  $n$  linearly independent vectors in  $\mathbb{R}^n$  is automatically a basis.

## Change of Basis

### Definition (Transition Matrix)

Let  $\mathcal{B}_1 = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$  and  $\mathcal{B}_2 = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$  be two ordered bases for  $\mathbb{R}^n$ . The *transition matrix*  $P$  from  $\mathcal{B}_1$  to  $\mathcal{B}_2$  has as its  $i$ -th column the coordinates of  $\mathbf{u}_i$  expressed in the basis  $\mathcal{B}_2$ . If  $[\mathbf{x}]_{\mathcal{B}_1}$  denotes the coordinate vector of  $\mathbf{x}$  in basis  $\mathcal{B}_1$ , then

$$[\mathbf{x}]_{\mathcal{B}_2} = P [\mathbf{x}]_{\mathcal{B}_1}.$$

The inverse  $P^{-1}$  converts in the opposite direction.

*Remark* (NURBS connection: Bernstein basis as a basis for polynomials). The Bernstein polynomials  $\{B_{0,n}(u), B_{1,n}(u), \dots, B_{n,n}(u)\}$  form a basis for the vector space of polynomials of degree  $\leq n$ . So does the monomial (power) basis  $\{1, u, u^2, \dots, u^n\}$ . The matrix  $M$  from Piegls & Tiller Exercise 1.14 is precisely the transition matrix converting coordinates in the monomial basis to coordinates in the Bernstein basis.  $M^{-1}$  converts the other direction. This is why the Bézier curve  $C(u) = [u^i]^T M [P_i]$  can be read as: “express the Bernstein basis functions in the monomial basis via  $M$ , then multiply by the control points.”

## 6.4 Orthogonality

## 6.5 Determinants

### Determinants and Cross Product — Quick Reference

|                         |                                                                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $2 \times 2$ <b>det</b> | $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$                                                                                               |
| $3 \times 3$ <b>det</b> | Cofactor expansion along any row or column.                                                                                                                  |
| <b>Cross product</b>    | $\mathbf{v} \times \mathbf{w}$ : vector orthogonal to both $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^3$ , computed via $3 \times 3$ determinant mnemonic. |
| <b>Magnitude</b>        | $\ \mathbf{v} \times \mathbf{w}\  = \ \mathbf{v}\  \ \mathbf{w}\  \sin \theta$ ; equals area of parallelogram.                                               |
| <b>Direction</b>        | Right-hand rule; $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$ .                                                                          |

### 6.5.1 Determinants and the Cross Product

#### The $3 \times 3$ Determinant

##### Definition ( $3 \times 3$ Determinant via Cofactor Expansion)

For  $A \in \mathbb{R}^{3 \times 3}$ , expanding along the first row:

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}).$$

The signs alternate:  $+, -, +$  along the first row.

#### The Cross Product

##### Definition (Cross Product)

For  $\mathbf{v} = (v_1, v_2, v_3)$  and  $\mathbf{w} = (w_1, w_2, w_3)$  in  $\mathbb{R}^3$ , the *cross product* is

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = (v_2w_3 - v_3w_2, \quad v_3w_1 - v_1w_3, \quad v_1w_2 - v_2w_1).$$

### Theorem (Properties of the Cross Product)

For  $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^3$  and  $c \in \mathbb{R}$ :

1.  $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$  (anticommutativity)
2.  $\mathbf{v} \times (\mathbf{w} + \mathbf{x}) = \mathbf{v} \times \mathbf{w} + \mathbf{v} \times \mathbf{x}$  (distributivity)
3.  $\mathbf{v} \times \mathbf{v} = \mathbf{0}$
4.  $(c\mathbf{v}) \times \mathbf{w} = c(\mathbf{v} \times \mathbf{w})$
5.  $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$  and  $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w}) = 0$  (orthogonality)
6.  $\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\| \|\mathbf{w}\| \sin \theta$  (magnitude = parallelogram area)

*Remark* (NURBS connection: surface normal vectors). The unit normal to a parametric surface at a point  $(u, v)$  is

$$N(u, v) = \frac{S_u \times S_v}{\|S_u \times S_v\|},$$

where  $S_u$  and  $S_v$  are the partial derivative vectors of the surface with respect to the parameters. The cross product gives the direction orthogonal to both tangent directions; dividing by the norm produces the unit normal. This appears in Piegl & Tiller §1.1 and is used throughout whenever normal vectors are needed (shading, offset surfaces, etc.).

*Remark* (Mnemonic for the cross product). Write the entries of  $\mathbf{v}$  and  $\mathbf{w}$  each twice in a  $2 \times 6$  array. Ignore the first and last columns. Draw three “X” shapes between columns 2–3, 3–4, and 4–5. Each “X” contributes one entry of  $\mathbf{v} \times \mathbf{w}$ : add along forward diagonals, subtract along backward diagonals.

## 6.6 Eigenvalues

## 6.7 Singular Value Decomposition

## 6.8 Linear Transformations

### Standard Matrix — Quick Reference

|                        |                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Standard matrix</b> | $[T] = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)] \in \mathbb{R}^{m \times n}$ . |
| <b>How to build it</b> | Evaluate $T$ on each standard basis vector; place results as columns.                                         |
| <b>Uniqueness</b>      | $[T]$ is the <i>unique</i> matrix with $T(\mathbf{v}) = [T]\mathbf{v}$ for all $\mathbf{v}$ .                 |
| <b>Composition</b>     | $[S \circ T] = [S][T]$ .                                                                                      |
| <b>Change of basis</b> | Same $T$ , different bases $\Rightarrow$ different matrix; related by $B = P^{-1}AP$ .                        |

### Linear Transformations — Quick Reference

|                        |                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b>      | $T(c\mathbf{v} + d\mathbf{w}) = cT(\mathbf{v}) + dT(\mathbf{w})$ for all $\mathbf{v}, \mathbf{w}$ and scalars $c, d$ . |
| <b>Origin fixed</b>    | Always $T(\mathbf{0}) = \mathbf{0}$ ; translation is <i>not</i> linear.                                                |
| <b>Standard matrix</b> | $[T] = [T(\mathbf{e}_1) \mid \cdots \mid T(\mathbf{e}_n)]$ ; then $T(\mathbf{v}) = [T]\mathbf{v}$ .                    |
| <b>Composition</b>     | $S \circ T$ has matrix $[S][T]$ .                                                                                      |
| <b>Affine map</b>      | $\Phi(\mathbf{r}) = A\mathbf{r} + \mathbf{v}$ : linear part $A$ plus translation $\mathbf{v}$ .                        |
| <b>Invariance</b>      | B-spline curves: apply affine map to control points only.                                                              |

### 6.8.1 The Idea of a Linear Transformation

#### Definition

##### Definition (Linear Transformation)

A function  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is a *linear transformation* if

$$T(c\mathbf{v} + d\mathbf{w}) = cT(\mathbf{v}) + dT(\mathbf{w})$$

for all  $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$  and all  $c, d \in \mathbb{R}$ .

Equivalently,  $T$  must satisfy both:

1.  $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$  (preserves addition), and
2.  $T(c\mathbf{v}) = cT(\mathbf{v})$  (preserves scalar multiplication).

*Remark* (The origin is always fixed). Every linear transformation satisfies  $T(\mathbf{0}) = \mathbf{0}$ . Therefore, *translation* (shifting by a nonzero vector) is not linear. The four geometric operations that *are* linear: rotation about the origin, scaling, reflection, and projection.

## Geometric Catalog in $\mathbb{R}^2$

### Standard Planar Transformations

|                                         |                                                                                                                      |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Scaling</b>                          | $\begin{pmatrix} c_1 & 0 \\ 0 & c_2 \end{pmatrix}$ : scales $x$ by $c_1$ , $y$ by $c_2$ .                            |
| <b>Rotation by <math>\theta</math></b>  | $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ : rotates counter-clockwise. |
| <b>Reflection (<math>y = x</math>)</b>  | $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ : swaps $x$ and $y$ coordinates.                                      |
| <b>Projection (<math>x</math>-axis)</b> | $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ : drops the $y$ component.                                            |

*Remark* (Composition is matrix multiplication). Applying  $T_1$  then  $T_2$  has matrix  $[T_2][T_1]$ . A sequence of geometric operations on a model (scale, then rotate, then translate) corresponds to multiplying their matrices together in that order.

## Affine Transformations

### Definition (Affine Transformation)

An *affine transformation*  $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}^n$  has the form

$$\Phi(\mathbf{r}) = A\mathbf{r} + \mathbf{v},$$

where  $A \in \mathbb{R}^{n \times n}$  is the linear part and  $\mathbf{v} \in \mathbb{R}^n$  is a translation vector. An affine transformation is a linear transformation when  $\mathbf{v} = \mathbf{0}$ .

*Remark* (Affine invariance of B-spline curves — NURBS Property P3.4). An affine transformation  $\Phi(\mathbf{r}) = A\mathbf{r} + \mathbf{v}$  applied to a B-spline curve  $C(u) = \sum_{i=0}^n N_{i,p}(u)P_i$  gives

$$\Phi(C(u)) = \sum_{i=0}^n N_{i,p}(u) \Phi(P_i).$$

The transformation can be applied to the curve by applying it to each control point individually. This works because the basis functions sum to 1 (partition of unity):

$$\Phi(C(u)) = A\left(\sum N_{i,p}P_i\right) + \mathbf{v} = \sum N_{i,p}(AP_i) + \underbrace{\left(\sum N_{i,p}\right)}_{=1}\mathbf{v} = \sum N_{i,p}(AP_i + \mathbf{v}).$$

This means rotating, translating, or scaling a B-spline curve costs only  $n+1$  affine operations on control points, regardless of how many points are evaluated on the curve.

## 6.8.2 The Matrix of a Linear Transformation

### The Standard Matrix

#### Theorem (Standard Matrix of a Linear Transformation)

A function  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is a linear transformation if and only if there exists a unique matrix  $[T] \in \mathbb{R}^{m \times n}$  such that

$$T(\mathbf{v}) = [T]\mathbf{v} \quad \text{for all } \mathbf{v} \in \mathbb{R}^n.$$

The *standard matrix*  $[T]$  is constructed by placing the images of the standard basis vectors as columns:

$$[T] = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)].$$

*Remark* (Why this works). Since every  $\mathbf{v} = v_1\mathbf{e}_1 + \cdots + v_n\mathbf{e}_n$ , linearity gives

$$T(\mathbf{v}) = v_1T(\mathbf{e}_1) + \cdots + v_nT(\mathbf{e}_n) = [T]\mathbf{v}.$$

Knowing  $T$  on the basis vectors determines  $T$  on the entire space.

### Matrix Under Change of Basis

#### Definition (Matrix Relative to a Basis)

Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$  be a linear transformation and let  $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$  be an ordered basis. The *matrix of  $T$  relative to  $\mathcal{B}$* , denoted  $[T]_{\mathcal{B}}$ , has as its  $j$ -th column the coordinates of  $T(\mathbf{b}_j)$  expressed in the basis  $\mathcal{B}$ .

If  $P$  is the transition matrix whose columns are  $\mathbf{b}_1, \dots, \mathbf{b}_n$  expressed in the standard basis, then

$$[T]_{\mathcal{B}} = P^{-1}[T]P.$$

Two matrices related by  $B = P^{-1}AP$  are called *similar*.

*Remark* (NURBS connection: Bézier curve in matrix form). The Bézier curve has the matrix representation

$$C(u) = [u^i]^T M [P_i],$$

where:

- $[u^i]^T = (1, u, u^2, \dots, u^n)$  is the monomial basis vector,
- $[P_i]$  is the column vector of control points,
- $M$  is the  $(n+1) \times (n+1)$  matrix that encodes the Bernstein polynomials in the monomial basis.

The matrix  $M$  is the standard matrix of the linear transformation that converts from Bernstein basis coordinates to monomial basis coordinates. Setting  $[\mathbf{a}_i] = M[P_i]$  gives the power basis representation of the same curve (so the Bézier coefficients  $[P_i]$  are the coordinates

of the curve in the Bernstein basis, and  $[\mathbf{a}_i]$  are the coordinates in the monomial basis). Converting between the two is exactly the change-of-basis operation  $P^{-1}AP$  applied to the coefficient vectors.

## 6.9 Complex Vectors and Matrices

## 6.10 Applications

## 6.11 Numerical Linear Algebra

## 6.12 Probability and Statistics

# Proofs

# Chapter 7

## Fourier Analysis



### Proofs

# Chapter 8

## Ordinary Differential Equations



## 8.1 Breadcrumb

### Where This Chapter Lives

Calculus → **Ordinary Differential Equations** → Dynamical Systems / Numerical Methods / PDE

**Why this chapter belongs here.** Ordinary differential equations are where derivatives stop being only local rates and start governing whole families of evolving states. Once differentiation is in hand, equations such as

$$y'(t) = f(t, y(t))$$

become the natural language for motion, growth, decay, forcing, and feedback.

**What this chapter will gather.** The eventual ODE chapter is the right home for:

- first-order equations and separable models;
- linear equations and integrating factors;
- existence and uniqueness ideas;
- systems of ODEs and phase portraits;
- stability, equilibrium, and qualitative behavior;
- numerical methods such as Euler and Runge–Kutta schemes.

**Why this is currently a breadcrumb.** The surrounding curriculum is still building the supporting architecture: calculus, linear algebra, and numerical analysis. ODEs sit at the meeting point of all three, so this file marks the destination clearly even before the chapter is fully written.

## 8.2 Models

### Definition

**Definition 8.2.1** (Differential Equation). A **differential equation** is an equation whose unknown is a function and whose terms involve one or more derivatives of that function.

For an unknown function  $y = y(t)$ , a differential equation may be written in the form

$$F(t, y, y', y'', \dots, y^{(n)}) = 0.$$

Here  $t$  is the independent variable,  $y(t)$  is the unknown function, and

$$y', y'', \dots, y^{(n)}$$

are derivatives of  $y$ .

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). At the computational level, a differential equation describes how a quantity changes. Instead of giving the function  $y(t)$  directly, it gives a relation involving the rate of change of  $y(t)$ , and possibly higher-order rates of change.

*Example* (A First Differential Equation). The equation

$$y' = 3y$$

is a differential equation because it relates the unknown function  $y$  to its derivative  $y'$ .

A solution is

$$y(t) = Ce^{3t},$$

where  $C$  is a constant. Indeed,

$$y'(t) = 3Ce^{3t} = 3y(t).$$

### Definition

**Definition 8.2.2** (Order of a Differential Equation). The **order** of a differential equation is the highest derivative of the unknown function that appears in the equation.

For example, if an equation involving an unknown function  $y = y(t)$  can be written in the form

$$F(t, y, y', y'', \dots, y^{(n)}) = 0$$

and  $y^{(n)}$  is the highest derivative that appears, then the differential equation has **order**  $n$ .

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Example* (Orders of Differential Equations). The equation

$$y' = 3y$$

is first order, since the highest derivative appearing is  $y'$ .

The equation

$$y'' + 4y' + 5y = 0$$

is second order, since the highest derivative appearing is  $y''$ .

The equation

$$y^{(4)} - ty'' + y = 0$$

is fourth order, since the highest derivative appearing is  $y^{(4)}$ .

### Definition

**Definition 8.2.3** (Solution of a Differential Equation). A **solution** of a differential equation is a function that satisfies the equation on a specified interval. More precisely, if a differential equation is written in the form

$$F(t, y, y', y'', \dots, y^{(n)}) = 0,$$

then a solution on an interval  $I$  is a function  $y : I \rightarrow \mathbb{R}$  such that  $y$  has all derivatives required by the equation and

$$F(t, y(t), y'(t), y''(t), \dots, y^{(n)}(t)) = 0$$

for every  $t \in I$ .

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). A solution is not a number; it is a function. The function solves the differential equation when substituting the function and its derivatives into the equation makes the equation true for every point in the interval under consideration.

*Example* (Checking a Solution). The function

$$y(t) = Ce^{3t}$$

is a solution of

$$y' = 3y$$

on  $\mathbb{R}$ , because

$$y'(t) = 3Ce^{3t} = 3y(t)$$

for every  $t \in \mathbb{R}$ .

### Definition

**Definition 8.2.4** (Autonomous Differential Equation). An **autonomous differential equation** is a differential equation in which the independent variable does not appear explicitly.

For a first-order ordinary differential equation, an autonomous equation has the form

$$y' = f(y),$$

where the rate of change of  $y$  depends only on the current value of  $y$ , not directly on time  $t$ .

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). An autonomous differential equation describes a system whose rule of evolution does not change with time. If the system is in the same state  $y$  at two different times, then the differential equation assigns the same rate of change at both times.

*Example* (Autonomous Equations). The equation

$$y' = 3y$$

is autonomous, since the right-hand side depends only on  $y$ .

The equation

$$y' = t + y$$

is not autonomous, since the independent variable  $t$  appears explicitly.

### Definition

**Definition 8.2.5** (Non-Autonomous Differential Equation). A **non-autonomous differential equation** is a differential equation in which the independent variable appears explicitly.

For a first-order ordinary differential equation, a non-autonomous equation has the form

$$y' = f(t, y),$$

where the rate of change of  $y$  may depend both on the current value of  $y$  and on the independent variable  $t$ .

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). A non-autonomous differential equation describes a system whose rule of evolution may change over time. Even if the system has the same state  $y$  at two different times, the differential equation may assign different rates of change because the value of  $t$  may be different.

*Example* (Non-Autonomous Equations). The equation

$$y' = t + y$$

is non-autonomous, since the independent variable  $t$  appears explicitly.

The equation

$$y' = 3y$$

is autonomous, not non-autonomous, since the right-hand side depends only on  $y$ .

### Definition

**Definition 8.2.6** (Linear Differential Equation). A differential equation is called **linear** if the unknown function and its derivatives appear only to the first power and are not multiplied together.

An  $n$ th-order linear ordinary differential equation for an unknown function  $y = y(t)$  has the form

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = g(t),$$

where the coefficient functions

$$a_0(t), a_1(t), \dots, a_n(t)$$

and the forcing term  $g(t)$  are given functions of the independent variable  $t$ .

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). Linearity means that the equation is linear in the unknown function  $y$  and its derivatives. The coefficients may depend on  $t$ , but they may not depend on  $y$  or its derivatives. Thus terms such as  $y^2$ ,  $yy'$ ,  $\sin(y)$ , and  $(y')^2$  make an equation nonlinear.

*Example* (Linear and Nonlinear Equations). The equation

$$y'' + 3ty' + 2y = \sin t$$

is linear.

The equation

$$y'' + y^2 = 0$$

is nonlinear because it contains the term  $y^2$ .

The equation

$$y' + yy' = t$$

is nonlinear because it contains the product  $yy'$ .

Ordinary differential equations are often used to model physical phenomena because many natural systems are governed by relationships between a quantity and its rate of change. In such models, the unknown function represents the state of the system, while the differential equation describes how that state evolves over time. For example, population models describe how a population changes, cooling models describe how temperature changes, motion models describe how position and velocity change, and circuit models describe how current or voltage changes. In each case, the differential equation does not usually give the quantity directly; instead, it gives a rule for how the quantity changes.

### Definition

**Definition 8.2.7** (Malthusian Population Growth Model). The **Malthusian population growth model** assumes that the rate of change of a population is proportional to the current population.

If  $P(t)$  denotes the population at time  $t$ , then the model is

$$P'(t) = kP(t),$$

where  $k$  is a constant called the **growth rate** or **proportionality constant**.

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). The model says that population growth depends only on the current population size. If  $k > 0$ , the population grows exponentially. If  $k < 0$ , the population decays exponentially. If  $k = 0$ , the population remains constant.

*Example* (General Solution). The general solution of

$$P'(t) = kP(t)$$

is

$$P(t) = Ce^{kt},$$

where  $C$  is a constant. If the initial population is  $P(0) = P_0$ , then

$$P(t) = P_0e^{kt}.$$

### Definition

**Definition 8.2.8** (Decay). A quantity is said to undergo **decay** if it decreases over time. In a differential equation model, if  $Q(t)$  denotes the quantity at time  $t$ , then decay is often modeled by an equation of the form

$$Q'(t) = -kQ(t),$$

where  $k > 0$  is a constant.

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). The equation

$$Q'(t) = -kQ(t)$$

says that the rate of change of the quantity is proportional to the current amount, but with a negative sign. Thus larger amounts decay faster in absolute size, while smaller amounts decay more slowly.

*Example* (Exponential Decay). The general solution of

$$Q'(t) = -kQ(t)$$

is

$$Q(t) = Ce^{-kt}.$$

If  $Q(0) = Q_0$ , then

$$Q(t) = Q_0e^{-kt}.$$

For  $k > 0$ , this function decreases toward 0 as  $t$  increases.

### Definition

**Definition 8.2.9** (Newton's Law of Cooling). **Newton's Law of Cooling** states that the rate at which an object's temperature changes is proportional to the difference between the object's temperature and the surrounding ambient temperature.

If  $T(t)$  denotes the temperature of the object at time  $t$ , and  $T_a$  denotes the ambient temperature, then the model is

$$T'(t) = -k(T(t) - T_a),$$

where  $k > 0$  is a constant called the **cooling constant**.

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). The quantity

$$T(t) - T_a$$

measures how far the object's temperature is from the ambient temperature. Newton's Law of Cooling says that the larger this temperature difference is, the faster the object cools or warms toward the ambient temperature.

*Example* (General Solution). The general solution of

$$T'(t) = -k(T(t) - T_a)$$

is

$$T(t) = T_a + Ce^{-kt},$$

where  $C$  is a constant. If  $T(0) = T_0$ , then

$$T(t) = T_a + (T_0 - T_a)e^{-kt}.$$

### Definition

**Definition 8.2.10** (General Solution). A **general solution** of a differential equation is a family of solutions containing one or more arbitrary constants.

For an  $n$ th-order ordinary differential equation, the general solution typically contains  $n$  arbitrary constants:

$$y(t) = \Phi(t, C_1, C_2, \dots, C_n).$$

Each choice of the constants  $C_1, C_2, \dots, C_n$  gives a particular solution of the differential equation.

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). The general solution represents the full family of solutions produced by solving the differential equation. Initial conditions are then used to choose specific values of the arbitrary constants, producing a particular solution.

*Example* (General Solution). The differential equation

$$y' = 3y$$

has general solution

$$y(t) = Ce^{3t},$$

where  $C$  is an arbitrary constant. If an initial condition such as

$$y(0) = 5$$

is imposed, then  $C = 5$ , and the particular solution is

$$y(t) = 5e^{3t}.$$

### Definition

**Definition 8.2.11** (Family of Solutions). A **family of solutions** of a differential equation is a collection of solution functions described by one or more parameters. For example, a family of solutions may be written in the form

$$y(t) = \Phi(t, C),$$

or more generally,

$$y(t) = \Phi(t, C_1, C_2, \dots, C_n),$$

where  $C, C_1, \dots, C_n$  are parameters. Each choice of the parameters gives one particular solution.

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). A family of solutions records many solutions at once. The parameters act as labels that distinguish one solution curve from another. When initial conditions are supplied, they usually determine the parameter values and select a single member of the family.

*Example* (A Family of Solutions). The differential equation

$$y' = 3y$$

has the family of solutions

$$y(t) = Ce^{3t},$$

where  $C \in \mathbb{R}$ . Each value of  $C$  gives a different solution.

### Definition

**Definition 8.2.12** (Initial Condition). An **initial condition** specifies the value of the unknown function at a particular value of the independent variable.

For a first-order differential equation involving an unknown function  $y = y(t)$ , an initial condition has the form

$$y(t_0) = y_0,$$

where  $t_0$  is the initial time and  $y_0$  is the initial value.

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). An initial condition selects one solution from a family of solutions. If a differential equation has general solution

$$y(t) = \Phi(t, C),$$

then the condition

$$y(t_0) = y_0$$

is used to determine the constant  $C$ .

*Example* (Initial Condition). The differential equation

$$y' = 3y$$

has general solution

$$y(t) = Ce^{3t}.$$

If the initial condition is

$$y(0) = 5,$$

then

$$5 = Ce^0 = C.$$

Thus the particular solution satisfying the initial condition is

$$y(t) = 5e^{3t}.$$

### Definition

**Definition 8.2.13** (Initial Value Problem). An **initial value problem**, or **IVP**, is a differential equation together with one or more initial conditions.

For a first-order differential equation, an initial value problem has the form

$$y' = f(t, y), \quad y(t_0) = y_0.$$

The differential equation describes how the function changes, while the initial condition specifies the value of the solution at the initial time  $t_0$ .

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). An initial value problem asks for the particular solution of a differential equation that passes through a specified point. Geometrically, the condition

$$y(t_0) = y_0$$

means that the solution curve must pass through the point

$$(t_0, y_0).$$

*Example* (Initial Value Problem). Consider the initial value problem

$$y' = 3y, \quad y(0) = 5.$$

The differential equation has general solution

$$y(t) = Ce^{3t}.$$

Using the initial condition,

$$5 = y(0) = Ce^0 = C.$$

Therefore the solution of the initial value problem is

$$y(t) = 5e^{3t}.$$

### Definition

**Definition 8.2.14** (Existence). An initial value problem is said to have **existence** if there is at least one function that satisfies both the differential equation and the initial condition.

For a first-order initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0,$$

existence means that there is a function  $y(t)$ , defined on some interval containing  $t_0$ , such that

$$y'(t) = f(t, y(t))$$

and

$$y(t_0) = y_0.$$

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

### Definition

**Definition 8.2.15** (Uniqueness). An initial value problem is said to have **uniqueness** if it has at most one solution on the interval under consideration.

For a first-order initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0,$$

uniqueness means that if  $y_1(t)$  and  $y_2(t)$  are both solutions on the same interval  $I$  containing  $t_0$ , then

$$y_1(t) = y_2(t)$$

for every  $t \in I$ .

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

### Definition

**Definition 8.2.16** (Interval of Existence). An **interval of existence** for a solution of an initial value problem is an interval on which the solution is defined and satisfies the differential equation.

For an initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0,$$

an interval  $I$  is an interval of existence if  $t_0 \in I$  and there is a solution  $y : I \rightarrow \mathbb{R}$  such that

$$y'(t) = f(t, y(t))$$

for every  $t \in I$ , and

$$y(t_0) = y_0.$$

*Remark* (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

*Remark* (Interpretation). Existence asks whether at least one solution passes through the initial point. Uniqueness asks whether only one solution passes through that point. The interval of existence records how long the solution is valid.

*Example* (Existence and Uniqueness). The initial value problem

$$y' = 3y, \quad y(0) = 5$$

has the solution

$$y(t) = 5e^{3t}.$$

This solution exists and is unique on  $\mathbb{R}$ .

The interval of existence is

$$(-\infty, \infty).$$

# Proofs

# Chapter 9

## Partial Differential Equations



### 9.1 Partial Differential Equations

*Remark* (Planned content). Active mathematical content has not yet been added.

### Proofs

# Chapter 10

## Computational Probability



### 10.1 Computational Probability

*Remark* (Planned content). Active mathematical content has not yet been added.

### Proofs

# Bibliography